Architecture-Constrained Likelihood Geometry for Detector Reconstruction:

A Framework and a Kronecker-Factored Case Study

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Abstract

Paper 1 established the likelihood geometry of detector reconstruction: under Gaussian noise with covariance Σ, the precision operator Σ^{-1} is the metric of the problem, and optimal filtering, EMPCA, and tied noise-aware linear autoencoders are all Σ^{-1}-orthogonal projections onto subspaces drawn from the full Grassmannian. This paper asks the next question: *once the residual geometry is fixed by the likelihood, which architecture priors are justified, and under what training coverage are the corresponding physical directions identifiable?* We answer it with a single framework: every reconstruction method is described by the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}} \to P_{\text{target}})$ — the residual geometry, the admissible map class, and the training coverage with any target-domain adaptation — with the loss fixed by Σ^{-1} and the architecture and coverage as separate, equally necessary choices. Geometry alone determines neither what a model can represent nor what it can identify.

The concrete, validated case study of the framework is the Noise-Factored Projection Autoencoder (NFPA), the simplest controlled architecture-constrained model: a Kronecker-constrained tied linear autoencoder that restricts the Paper 1 bridge objective from the full Grassmannian $\mathrm{Gr}(k, CT)$ to the product geometry $\mathrm{Gr}(k_c, C) \times \mathrm{Gr}(k_t, T)$. NFPA changes the admissible projector class, not the likelihood. We prove a restricted bridge theorem — NFPA solves the EMPCA/linear-autoencoder objective restricted to Kronecker-separable channel-time

subspaces, with equality to EMPCA only when an EMPCA minimizer is itself separable — and show that rank-one NFPA recovers the optimal filter under a separable whitened template. In a controlled synthetic experiment the metric reversal of Paper 1 persists under the Kronecker architecture: NFPA improves the likelihood-aligned residual and captures the dominant EMPCA direction, while the MSE-trained factored model wins only in raw Euclidean error and learns a nearly orthogonal subspace.

The same tuple organizes the rest of the paper. A short external case study (U-Net denoising for a Xenon TPC) *illustrates* the geometry/architecture/ coverage separation on a nonlinear pipeline, and noise-aware VAE and Transformer models are presented as framework-derived extensions — each with a stated prior, coverage requirement, failure mode, and concrete diagnostics such as pullback geometry, time–frequency ablations, and local-linear Jacobian tests — rather than as completed contributions. The result is a framework, with one validated Kronecker-factored case study, from which a concrete experimental program follows directly.

1 Introduction

Modern low-threshold detectors reconstruct small signals from noisy, multi-channel time series rather than counting threshold crossings. Paper 1 (Wong, 2026) answered one half of this problem — the metric. This paper asks the other half:

> *Once the residual geometry is fixed by the detector likelihood, which architecture priors are justified, and under what training coverage are the corresponding physical directions identifiable?*

Paper 1 answered the metric question; Paper 2 answers the architecture-and-coverage question.

What Paper 1 fixed. Given detector traces corrupted by Gaussian noise with known covariance Σ, the maximum-likelihood reconstruction loss is the Mahalanobis loss,

$$\|x - \hat{x}\|^2_{\Sigma^{-1}} = (x - \hat{x})^\dagger \Sigma^{-1} (x - \hat{x}), \tag{1}$$

the correct linear encoder is the Σ^{-1}-adjoint of the decoder, and optimal filtering, EMPCA, and tied noise-aware linear autoencoders are unified as Σ^{-1}-orthogonal projections. In whitened coordinates $\tilde{x} = \Sigma^{-1/2}x$ this is ordinary Euclidean projection; in detector coordinates it is a noise-induced metric geometry, with the optimization running over the full Grassmannian of rank-k subspaces. The metric is settled; the *map class* and the *training distribution* are not.

Why the metric is not enough. Two facts motivate the rest of the paper. First, two models trained with the *same* Mahalanobis loss can behave very differently, because their architectures encode different priors — locality, factorization, graph structure, attention — and therefore admit different reconstruction maps. Second, even an arbitrarily expressive architecture cannot recover variation along a latent direction that the training distribution never excites: if the simulation fixes the arrival time or the interaction position, the loss carries no information about those directions. Scientific reconstruction therefore requires three coupled choices, which we write as the tuple

$$\big(\Sigma^{-1},\ \mathcal{F},\ P_{\text{train}} \to P_{\text{target}}\big), \tag{2}$$

whose three entries are the residual geometry Σ^{-1} (how residuals are weighted), the architecture class $\mathcal{F}$ (which maps are admissible), and the coverage-and-adaptation $P_{\text{train}} \to P_{\text{target}}$ (which directions are identifiable). We use this tuple repeatedly throughout the paper. The arrow records that the training distribution may differ from the deployment distribution and may be partially repaired by a target-domain adaptation protocol (Section 7); absent such a protocol, $P_{\text{target}} = P_{\text{train}}$. We state results about reconstruction in terms of *inductive bias*, *approximation class*, and *identifiability*, rather than the vague claim that a model "learns the physics." Table 1 fixes the division of labor.

To keep the three ingredients separate we carry the decomposition

$$\Omega \ \longrightarrow\ \mathcal{X}(\Omega) \ \longrightarrow\ \mathcal{F} \ \longrightarrow\ \mathcal{M} \tag{3}$$

throughout: the detector domain Ω (time bins, channels, or their product) indexes the measurement; the signal space $\mathcal{X}(\Omega)$ carries the noise metric Σ^{-1}; the hypothesis class $\mathcal{F}$ is selected by the architecture; and the event manifold $\mathcal{M} = \{s(z)\}$ is the family of clean signals generated by physical parameters $z = (E, t_0, r, \phi, \gamma, \ldots)$, whose coverage by P_{train} controls identifiability.

Table 1: The three coupled ingredients of architecture-constrained likelihood geometry, written as the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}})$. The loss is fixed by Σ^{-1} and is architecture-neutral; the architecture and the training distribution are separate, equally necessary choices.

Ingredient	Question it answers	Fixed by	Treated in
Likelihood geometry Σ^{-1}	How should residuals be weighted?	Noise covariance	Paper 1; recap Sec. 2
Architecture class $\mathcal{F}$	Which maps can the model represent?	Model + detector domain	Secs. 3, 4, 7; Apps. B, C
Coverage & adaptation $P_{\text{train}} \to P_{\text{target}}$	Which directions are identifiable?	Training distribution (+ adaptation)	Sec. 2.6; Sec. 8

NFPA: the first worked example. The concrete, validated instance of the framework is the Noise-Factored Projection Autoencoder (NFPA), the simplest controlled architecture-constrained likelihood-geometry model. In the tuple language its residual geometry is the unchanged Mahalanobis/whitened loss; its architecture prior is a Kronecker-separable channel-time subspace; its coverage requirement is variation along the relevant channel and temporal modes; and its failure mode is that nonseparable signal modes cannot be represented without approximation error. Formally, Paper 1's bridge theorem optimizes over the full Grassmannian $\mathrm{Gr}(k, CT)$; NFPA restricts this to the product geometry $\mathrm{Gr}(k_c, C) \times \mathrm{Gr}(k_t, T)$, embedded into $\mathrm{Gr}(k_c k_t, CT)$ through $\mathrm{range}(U_c) \otimes \mathrm{range}(U_t)$. NFPA changes the admissible projector class, not the likelihood; the restricted bridge theorem (Proposition 4.1) makes this precise, and we are explicit that NFPA equals EMPCA only when an EMPCA minimizer is itself separable.

Scope. Only the NFPA contribution is experimentally validated, and only in one controlled synthetic setting; we treat it as a proof-of-concept case study and specify the sweeps a submission would require (Section 5). The coverage condition — the third ingredient — is developed in Section 2.6 as a detector-design rule. The noise-aware VAE (Appendix B) and noise-aware Transformer (Appendix C) are framework-derived extensions: for each we

state the prior, the coverage it requires, its failure mode, and the minimal experiment that would test it, but we make no empirical claim. An external case study (Section 7) illustrates the same separation on a nonlinear pipeline.

Contributions.

1. We formulate detector reconstruction as the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}})$ and use it as a single decision procedure: the loss is fixed by Σ^{-1}, while the architecture and coverage are separate, equally necessary choices.

2. We derive detector-native architecture priors — temporal grid, channel graph, channel-time product, event-manifold pullback metric, gauge freedom — and tabulate each method by metric, prior, coverage, failure mode, and experiment (Table 2).

3. We introduce NFPA as the first worked example and prove the restricted bridge theorem and the rank-one NFPA/OF correspondence under a separable whitened template.

4. We state the coverage condition as a detector-design rule, with a first-order tangent-recovery proposition and a Fisher-rank diagnostic.

5. We validate the completed NFPA case study: metric reversal persists after imposing Kronecker structure, and NFPA captures the dominant likelihood-aligned EMPCA direction while MSE-trained factored models learn nearly orthogonal subspaces.

6. We turn the framework into a concrete experimental program (Section 8) and into framework-derived VAE and Transformer extensions, including time–frequency and local-linear diagnostics (Appendices B, C).

Status of claims. The claims in this draft deliberately have different statuses:

Claim	Current status
Mahalanobis loss is the Gaussian detector likelihood metric and remains a residual-level requirement beyond linear decoders	Proven in Paper 1; reused here
NFPA is the Paper 1 bridge objective restricted to Kronecker-separable channel-time subspaces	Proven here
Rank-one NFPA recovers OF under a separable whitened template	Proven here
NFPA improves the likelihood-aligned residual relative to the MSE-trained factored model	Demonstrated in one synthetic setting (proof-of-concept)
Coverage failure leaves latent directions unidentifiable even under the correct loss	Theoretical principle; experiment specified
Noise-aware VAE / prewhitened Transformer improve detector reconstruction	Framework-derived extensions; not yet tested

2 From Likelihood Geometry to Domain Geometry

This section separates three distinct sources of structure that the rest of the paper keeps apart: the *observation-space metric* selected by the Gaussian likelihood, the *domain symmetries* selected by the detector geometry, and the *architecture-induced admissible maps* selected by modeling assumptions. Following the geometric deep-learning viewpoint (Bronstein et al., 2021), we distinguish the observation-space metric selected by the likelihood from the domain symmetries selected by the detector geometry.

2.1 Paper 1 in one page

Let $x_i = s_i + n_i \in \mathbb{C}^d$ be a detector observation with $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$ and known $\Sigma \succ 0$. The Gaussian negative log likelihood differs from

$$(x_i - \hat{s}_i)^\dagger \Sigma^{-1} (x_i - \hat{s}_i) \tag{4}$$

only by constants independent of the reconstruction. Therefore Σ^{-1} defines the observation-space inner product

$$\langle u, v \rangle_{\Sigma^{-1}} = u^\dagger \Sigma^{-1} v. \tag{5}$$

Whitening sends $\tilde{x} = \Sigma^{-1/2} x$ and converts this geometry into the ordinary Euclidean geometry of $\tilde{x}$.

Paper 1 used this metric to unify three linear estimators. The optimal filter is a rank-one Σ^{-1}-orthogonal projection onto a known template. EMPCA is a learned rank-k Σ^{-1}-orthogonal projection. A tied noise-aware linear autoencoder with decoder P and encoder $P^\dagger \Sigma^{-1}$ minimizes the same objective as EMPCA:

$$\min_{P^\dagger \Sigma^{-1} P = I_k} \frac{1}{N} \sum_i \left\| x_i - P P^\dagger \Sigma^{-1} x_i \right\|_{\Sigma^{-1}}^2 . \tag{6}$$

Thus the full linear bridge theorem is exact: the minimizer of the noise-aware tied linear autoencoder is the EMPCA subspace.

2.2 Metric symmetry versus domain symmetry

The inner product $\langle u, v \rangle_{\Sigma^{-1}}$ defines the *noise-isometry group*

$$G_{\text{noise}} = \{ Q : Q^\dagger \Sigma^{-1} Q = \Sigma^{-1} \}. \tag{7}$$

In whitened coordinates, G_{noise} becomes the ordinary orthogonal (real case) or unitary (complex case) group. This symmetry is the reason the correct linear encoder is the Σ^{-1}-adjoint of the decoder and why tied weights are an adjointness condition rather than a heuristic regularizer: among quadratic residual losses, G_{noise}-invariance selects Σ^{-1} up to scale.

It is essential not to conflate this *metric symmetry* with *domain symmetry*. G_{noise} preserves the residual norm; it acts on observation space and knows nothing about the detector layout. It is not the same as time translation, channel relabeling, detector rotation, or graph automorphism, which act on the domain Ω and may (approximately) commute with the physical signal-generating process. In particular, prewhitening fixes the residual metric but does not by itself impose convolutional locality, graph equivariance, or deformation stability. Those inductive biases must be encoded by the architecture, and their validity is a property of the detector, not of the noise model.

2.3 What changes beyond full linear subspaces

The metric does not change when the decoder class changes. If the noise model is still $n \sim \mathcal{N}(0, \Sigma)$, then the likelihood still uses Σ^{-1}. What changes is the admissible reconstruction family. In full EMPCA, the admissible family is all rank-k linear subspaces. In NFPA, the admissible family is only separable channel-time subspaces. In a VAE, it is the image of a nonlinear decoder with a latent prior. In a Transformer, it is a sequence model acting on whitened tokens.

This is why the exact EMPCA/AE bridge belongs in Paper 1, while NFPA belongs here. NFPA is still linear, but it is not the unconstrained linear bridge. It adds architectural structure. The theorem is therefore not "NFPA equals EMPCA"; it is "NFPA equals the EMPCA objective restricted to Kronecker subspaces."

2.4 Domain, signal space, hypothesis class, manifold, and gauge

It is useful to keep five objects separate, following the geometric deep learning language of Bronstein et al. (Bronstein et al., 2021), as in the decomposition (3).

1. *Domain* Ω: the index set on which observations are sampled — time bins, readout channels, pixels, graph nodes, or products of these.

2. *Signal space* $\mathcal{X}(\Omega)$: detector traces, images, or functions on the domain, equipped here with the noise metric Σ^{-1}.

3. *Hypothesis class* $\mathcal{F}$: the family of reconstruction maps allowed by the chosen architecture.

4. *Latent/event manifold* $\mathcal{M}$: the family of clean signals $s(z)$ generated by physical parameters $z = (E, t_0, r, \phi, \gamma, \ldots)$.

5. *Gauge freedom*: non-identifiability of basis or latent coordinates under transformations that leave reconstructions invariant.

Paper 1 identifies the correct geometry on $\mathcal{X}(\Omega)$ for Gaussian detector noise. Paper 2 changes $\mathcal{F}$ while keeping that observation-space geometry fixed, and asks how the geometry of Ω and $\mathcal{M}$ should constrain $\mathcal{F}$. Domain symmetries

such as translation, graph relabeling, graph locality, or time-warping should be built into the architecture; the covariance Σ determines the likelihood and therefore the residual metric. This separation prevents the two roles from being conflated: prewhitening does not by itself supply a convolutional, recurrent, or graph inductive bias, and an expressive architecture does not by itself repair a misspecified isotropic loss.

2.5 The geometry-first design checklist

The practical design rule that transfers from Paper 1 is a checklist, not a single architecture:

1. *Identify the noise metric.* The covariance Σ defines a Riemannian metric Σ^{-1} on observation space.

2. *Set the loss.* Use the squared Mahalanobis distance, or Gaussian χ^2 loss, $(x - \hat{x})^\dagger \Sigma^{-1} (x - \hat{x})$; among quadratic residual losses, this is the form selected by G_{noise}-equivariance.

3. *For linear architectures, use the adjoint tie.* A decoder basis P should be paired with the encoder $P^\dagger \Sigma^{-1}$ and normalized by $P^\dagger \Sigma^{-1} P = I_k$, the Riesz–Frechet adjoint condition from Paper 1.

4. *For nonlinear architectures, whiten first.* Applying $\Sigma^{-1/2}$ as a fixed first layer moves the data into a space where the noise metric is Euclidean, after which standard nonlinear architectures can be used without rederiving the metric in each layer.

5. *For Transformers, prewhiten before attention.* Standard attention can then operate on whitened tokens, optionally supplemented by noise-spectral positional encodings derived from the precision operator.

2.6 Metric Geometry, Signal Coverage, and Architectural Selection

2.6.1 The sufficiency gap

The checklist above identifies the correct residual geometry, but it does not by itself solve nonlinear signal reconstruction. The Mahalanobis loss answers the question

> Given a candidate reconstruction error, how should that error be weighted under the detector noise model?

It does not answer the complementary questions

> Which physical factors were varied in training? Which signal-manifold directions were never excited? Among all reconstructions compatible with the likelihood on the observed support, which one should the model select?

The upgraded design rule is therefore:

$$\text{correct loss} \Longrightarrow \text{correct metric geometry on training support}, \tag{8}$$

whereas plausible physical reconstruction in the nonlinear setting requires

$$\text{correct loss} + \text{signal coverage} + \text{architectural inductive bias}. \tag{9}$$

This is a detector-reconstruction analogue of recent contrastive-learning theory showing that objective design alone does not determine latent recovery when sampling support is restricted (Zaliaduonis et al., 2026).

2.6.2 Signal-manifold coverage condition

Write the detector signal model as

$$x = g(z) + n, \qquad n \sim \mathcal{N}(0, \Sigma), \tag{10}$$

with physical latent variables

$$z = (E, t_0, r, \phi, c, \alpha, b, \eta), \tag{11}$$

representing deposited energy, arrival time, interaction position, channel-sharing pattern, pulse-shape deformation, baseline state, and detector noise state. Let P_Z^{target} denote the target distribution over physical latent variables and let P_Z^{train} denote the latent distribution induced by the simulation or calibration procedure used for training.

Definition 2.1 (Signal-manifold coverage condition)**.** *The training design satisfies the signal-manifold coverage condition on a target domain if the target latent support is contained in the training latent support,*

$$\operatorname{supp}(P_Z^{\text{target}}) \subseteq \operatorname{supp}(P_Z^{\text{train}}). \tag{12}$$

Equivalently, every latent region with nonzero target probability must remain reachable under the training mechanism. If training fixes a factor such as $t_0 = 0$ or $r = r_0$, the condition fails along that physical direction. When recovery is conditioned on a known context variable c, the corresponding local condition is

$$\operatorname{supp}(P_{Z|c}^{\text{target}}) \subseteq \operatorname{supp}(P_{Z|c}^{\text{train}}).$$

The condition is written in deliberate analogy with the diversity condition of Zaliaduonis et al. (Zaliaduonis et al., 2026), but its interpretation here is detector-native: the relevant support is not positive-pair support on a latent sphere, but coverage of the physical signal manifold under the calibrated noise model. In this draft it should be read as a design principle supplemented by the first-order identifiability statements below, not yet as a complete global recovery theorem.

2.6.3 Restricted coverage implies restricted tangent recovery

Proposition 2.1 (Restricted first-order recovery under coverage failure)**.** *Let $M = \{g(z) : z \in Z\}$ be the signal manifold, and let the training distribution vary only the latent coordinates indexed by $S \subseteq \{1, \dots, m\}$ while fixing the complementary coordinates S^c. In the population limit, the Mahalanobis reconstruction objective supplies first-order identifying information only along the tangent directions*

$$\operatorname{span}\left\{\Sigma^{-1/2}\frac{\partial g}{\partial z_j}(z) : j \in S\right\}. \tag{13}$$

The unvaried directions $\{\Sigma^{-1/2}\partial g/\partial z_j : j \in S^c\}$ are not excited by the training objective and therefore are not identified by loss geometry alone. Architecture, side information, or explicit physical priors may extrapolate into such directions, but that information does not come from the reconstruction loss on the restricted support.

Proof sketch. In whitened coordinates $\tilde{x} = \Sigma^{-1/2}x$, the loss is ordinary squared error on the whitened signal manifold $\tilde{g}(z) = \Sigma^{-1/2}g(z)$. If the training distribution fixes $z_j = z_j^\star$ for $j \in S^c$, then the population objective integrates only over variation in z_S. A local linearization gives

$$\tilde{g}(z_S, z_{S^c}^\star + \delta z_{S^c}) \approx \tilde{g}(z_S, z_{S^c}^\star) + \sum_{j \in S^c} \frac{\partial \tilde{g}}{\partial z_j}\delta z_j, \tag{14}$$

but the coefficients δz_j never vary in training. Consequently the population gradient of the reconstruction loss has no term that tests decoder sensitivity along those directions. The result is exact in the linear case and extends locally to the nonlinear case by the same Jacobian argument. □

Proposition 2.1 is intentionally a first-order identifiability statement, not yet a global uniqueness theorem. Its role is to make precise why the correct χ^2 geometry is necessary but insufficient when the training support omits relevant physical variation.

Remark 2.1 (Exact linear-support reduction)**.** *In the exactly linear case $\tilde{g}(z) = \tilde{g}_0 + \sum_{j=1}^{m} z_j q_j$ with whitened basis directions q_j, restricted coverage gives an exact subspace statement rather than only a local one. If the training distribution has zero variance in the coordinates z_j for $j \in S^c$, then*

$$\mathrm{Cov}(\tilde{g}(z)) = \sum_{j,\ell \in S} \mathrm{Cov}(z_j, z_\ell)\, q_j q_\ell^\top, \tag{15}$$

so every population linear Mahalanobis-optimal projector lies in $\mathrm{span}\{q_j : j \in S\}$*. This is the exact linear prototype for the nonlinear local-Jacobian argument above.*

Proposition 2.2 (Fisher-information rank diagnostic)**.** *Let $J_g(z)$ denote the Jacobian of the detector map g at latent state z, and define the local Fisher information matrix*

$$I(z) = J_g(z)^\top \Sigma^{-1} J_g(z). \tag{16}$$

If

$$\mathrm{rank}\, I(z) < \dim z, \tag{17}$$

then the physical latent variables are not locally identifiable at z under the Gaussian observation model. In particular, if the training distribution concentrates on a submanifold along which this rank drops, then local physical recovery is underdetermined regardless of decoder capacity.

Proof. For Gaussian observations with fixed covariance, the Fisher information for the mean parameterized by z is exactly $J_g(z)^\top \Sigma^{-1} J_g(z)$. If its rank is smaller than the latent dimension, then there exists a nonzero direction δz in the nullspace of $I(z)$, equivalently $J_g(z)\delta z = 0$ in the Σ^{-1} metric. Perturbations along that latent direction do not change the observation to first order, so standard local identifiability fails. □

2.6.4 Architecture as selection

Even when coverage is adequate, the likelihood generally leaves multiple reconstructions admissible on the observed support. Architecture selects among them. NFPA restricts the model class to $\mathrm{range}(U_c) \otimes \mathrm{range}(U_t)$ and therefore encodes a Kronecker channel-time factorization. A CNN autoencoder encodes temporal locality and approximate translation equivariance. A graph autoencoder can encode detector adjacency or a sparsified precision graph. A prewhitened Transformer allows adaptive long-range interactions between whitened tokens. None of these architectures replaces the likelihood metric; each is an inductive prior inside the same noise geometry.

2.6.5 Achievability versus selection

This distinction leads to a useful separation between achievability and selection. Coverage-corrected training schemes such as stratified simulation, importance weighting, or physically matched nuisance augmentation can make physically meaningful reconstruction achievable by restoring support along the relevant latent directions. They do not uniquely determine which achievable solution a flexible decoder will select. Architecture continues to play that selection role. This is the detector analogue of the achievability-without- uniqueness point emphasized for support-corrected InfoNCE by Zaliaduonis et al. (Zaliaduonis et al., 2026).

Remark 2.2 (Restricted-support distortion example)**.** *The support issue can be seen even without changing the metric. Suppose a whitened detector family admits local coordinates (u, v), where u denotes a well-sampled latent factor and v denotes a factor absent from training. If training data satisfy $v = 0$, then the reconstruction family*

$$\tilde{f}_\lambda(u, v) = \tilde{g}(u, \lambda v), \qquad \lambda > 0, \tag{18}$$

agrees exactly with the true map on training support for every λ. Hence all such maps are equally Mahalanobis-consistent in-support, even though they differ off-support whenever $\lambda \neq 1$. The point is not that this family is unique, but that restricted support leaves room for geometry-distorting solutions that the correct loss alone cannot exclude.

3 Architecture Priors for Detector Reconstruction

Once the residual metric Σ^{-1} is fixed by the likelihood, the remaining modeling freedom is the architecture class $\mathcal{F}$ — the family of reconstruction maps the model can express. We derive the admissible priors directly from detector structure: the *time grid* implies temporal locality and shift behavior; the *channel graph* implies adjacency, crosstalk, or covariance-informed coupling; the *channel* $\times$ *time product* implies separable or factorized models such as NFPA; the *event manifold* implies a pullback/Fisher metric for identifiability; and *gauge freedom* implies that raw latent coordinates must not be overinterpreted. These five correspond to the grids/groups/graphs/geodesics/gauges vocabulary of Bronstein et al. (Bronstein et al., 2021), which we use as supporting nomenclature rather than as the source of the priors; the full group/symmetry taxonomy and diagnostics are deferred to Appendix A. Throughout, the metric is held fixed; these structures constrain $\mathcal{F}$, not the likelihood.

3.1 Grids: temporal locality and shift structure

Waveforms live on a temporal grid. Temporal convolution is justified by local time structure and approximate translation equivariance: if a pulse shifts in time, the reconstructed trace should shift consistently; the energy estimate should be invariant to the shift; and the arrival-time estimate t_0 should transform equivariantly by addition, $t_0 \mapsto t_0 + \tau$. This motivates temporal CNN/ResNet baselines not because they are popular, but because they encode local time-grid priors and scale separation: local filters encode stability to small shifts and local pulse deformations, while residual and multiscale blocks combine fast rise features with slower decay and baseline-drift structure. A concrete failure mode follows directly: a temporal CNN whose receptive field is shorter than the separation between the pulse and the baseline window cannot use late-time baseline samples to constrain the amplitude, which is one reason multiscale or U-Net depth (as in Section 7) is preferred over a shallow local filter.

3.2 Graphs: channel geometry and product domains

Readout channels define a detector graph $G = (V, E)$, and the full domain is naturally a product:

$$\Omega = V_{\text{channel}} \times V_{\text{time}}. \tag{19}$$

Edges can encode physical adjacency, readout crosstalk, empirical noise covariance, or a sparsified precision pattern $|\Sigma_{ij}^{-1}|$. The three main architecture classes then differ in which graph they assume. CNNs assume a fixed local grid. GNNs assume an explicit detector graph and perform message passing along its edges. Transformers can be interpreted as *learned complete-graph message passing*; they assume no geometry, so geometry must be supplied — through positional encodings, attention biases, physical coordinates, noise spectra, or prewhitening. The concrete failure mode here is the dual of the grid case: imposing a 2D-grid convolution on electrodes that lack pixel adjacency (as in the Xenon TPC of Section 7) injects a false locality prior and can smear charge sharing between channels that are physically coupled but not grid neighbors.

An important caution: full channel permutation is usually *not* a physical symmetry. Channel 45 and channel 50 are not physically identical if their positions, calibrations, or noise differ. What is required is equivariance to *graph relabeling*: if node labels are reordered and the adjacency, coordinates, and covariance are reordered consistently, the model output should reorder consistently. This is the GNN-style symmetry, not a claim that sensors are exchangeable.

3.3 Geodesics: the event manifold and its pullback metric

The clean signal family is not just a cloud in $\mathbb{R}^{C\times T}$; it is an embedded manifold parameterized by physical event variables,

$$\mathcal{M} = \{s(z) : z = (E, t_0, r, \phi, \gamma, \ldots)\}. \tag{20}$$

Because residuals are measured with Σ^{-1}, the physically relevant metric induced on the event manifold is the pullback (equivalently, Fisher) metric

$$g_z = J_s(z)^\dagger \Sigma^{-1} J_s(z), \qquad J_s(z) = \frac{\partial s(z)}{\partial z}. \tag{21}$$

This metric identifies which latent/event directions are distinguishable under detector noise and which are degenerate (directions in the nullspace of g_z are unidentifiable to first order, the Fisher-rank diagnostic of Proposition 2.2), it defines geodesic distance between events in detector-distinguishability units, and it explains why Euclidean interpolation in a latent space can be physically wrong. For a learned nonlinear decoder $D_\theta(z)$, the same construction gives

$$g_z = J_{D_\theta}(z)^\dagger \Sigma^{-1} J_{D_\theta}(z), \tag{22}$$

which we present as the nonlinear analog of Paper 1's linear metric geometry: it is what "the subspace inherits the Σ^{-1} inner product" becomes when the decoder is nonlinear. We treat (22) as a definition and diagnostic tool, not an experimentally validated result; Appendix B develops the corresponding VAE diagnostics.

3.4 Gauges: what not to interpret

Every model here carries gauge freedom. EMPCA has an $O(k)$ (real) or $U(k)$ (complex) gauge, $U \mapsto UQ$, that leaves the projector unchanged; NFPA has the product gauge $U_c \mapsto U_c Q_c$, $U_t \mapsto U_t Q_t$ with $Q_c \in O(k_c)$, $Q_t \in O(k_t)$; and VAE latent coordinates are defined only up to reparameterization. The practical rule is therefore to interpret only gauge-invariant quantities — projectors, subspaces, reconstructions, pullback distances, and invariant/equivariant errors — and never raw latent coordinates. All subspace comparisons in this paper (principal angles, projection distances) are gauge-invariant by construction.

3.5 The framework tuple and a unified method table

The detector priors above turn each architecture into a specific instance of one decision procedure. We describe every method by the tuple

$$(\Sigma^{-1},\ \mathcal{F},\ P_{\text{train}}) \ =\ (\text{residual geometry, architecture class, training coverage}), \tag{23}$$

together with its characteristic *failure mode* and the *experiment* that would test it. Table 2 fills in this template for the methods of this paper. The residual-geometry column is deliberately almost constant — every method uses the same Σ^{-1} likelihood metric — which is precisely the point: the

methods differ in $\mathcal{F}$ and in the coverage they require, not in the loss. The architecture rows trace one hierarchy of increasingly weak structural priors,

$$\text{OF/EMPCA} \;\rightarrow\; \text{NFPA} \;\rightarrow\; \text{ResNet/U-Net} \;\rightarrow\; \text{Transformer}, \tag{24}$$

from linear noise-aware projection, through a separable channel-time product prior, to nonlinear temporal locality, to global adaptive attention. A final row is not an architecture at all but a *training strategy* — semi-supervised coverage correction — that acts on the $P_{\text{train}} \rightarrow P_{\text{target}}$ arrow rather than on $\mathcal{F}$. NFPA (Section 4) is the row this paper validates; ResNet/U-Net are developed in Section 6 and instantiated by the case study of Section 7; VAE and Transformer are framework-derived extensions in Appendices B and C.

4 Noise-Factored Projection Autoencoder

NFPA is the main completed contribution of this paper, and we frame it as the *first architecture-constrained bridge theorem*. Paper 1's bridge theorem optimizes over the full Grassmannian $\mathrm{Gr}(k, CT)$; NFPA restricts the same objective to the product geometry $\mathrm{Gr}(k_c, C) \times \mathrm{Gr}(k_t, T)$, embedded into $\mathrm{Gr}(k_c k_t, CT)$ through $\mathrm{range}(U_c) \otimes \mathrm{range}(U_t)$. In the 5G vocabulary of Section 3, NFPA imposes a product-domain prior (channel sharing and temporal pulse evolution factor) while leaving the likelihood metric untouched.

4.1 Data model and notation

Consider a multi-channel trace represented as a matrix

$$X_i \in \mathbb{R}^{C \times T}, \tag{25}$$

where C is the number of readout channels and T is the number of time samples. Let $Z_i = \Sigma^{-1/2} X_i$ denote the whitened representation. In the current implementation the whitening is applied in frequency space along the temporal dimension of each channel; the theory below is easier to write in matrix notation, but the vectorized form is equivalent.

NFPA assumes that the signal subspace factors into a channel subspace and a temporal subspace:

$$U_c \in \mathbb{R}^{C \times k_c}, \qquad U_t \in \mathbb{R}^{T \times k_t}, \qquad U_c^\top U_c = I_{k_c}, \quad U_t^\top U_t = I_{k_t}. \tag{26}$$

Table 2: The framework tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}} \to P_{\text{target}})$ applied to each method, with failure mode and the experiment that tests it. The residual geometry is shared; methods differ in admissible map class and in their coverage role. The last row is a training strategy, not an architecture. "App." refers to the appendices.

Method	Residual geometry	Architecture prior	Domain assumption	Coverage role	Failure mode	Experiment
OF / EMPCA	Σ^{-1} whitened	Fixed template (OF) or rank-k subspace (EMPCA)	Global linear subspace	Needs variation along each kept mode	Nonlinear / off-subspace shape	Rank sweep; metric reversal
NFPA	Σ^{-1} whitened	Kronecker channel $\times$ time subspace	Separable channel/time	Needs channel and temporal modes varied	Nonseparable modes (approx. error)	Angles vs EMPCA; nonseparable sweep
ResNet (temporal)	Σ^{-1} (whiten first)	Residual corrections to a baseline; temporal locality	Time grid	Needs shifts and deformations in train	Channel conv without a valid graph	Residual vs direct; restricted coverage
U-Net (temporal)	Σ^{-1} (whiten first)	Multiscale temporal conv with skips	Time grid (not channel grid)	Needs energy/shape range in train	Bottleneck loses sharp detail; false channel grid	Case study; coverage sweep
VAE	Σ^{-1} in ELBO recon. term	Nonlinear decoder + latent prior	Low-dim event manifold	Needs dense manifold sampling	Latent teleportation off-manifold	Pullback-geodesic vs isotropic (App. B)
Transformer	Σ^{-1} (prewhiten tokens)	Learned global pairwise attention	None (geometry supplied via PE)	Needs data to learn pairwise structure	Spurious long-range fits off-support	PE / graph ablations (App. C)
Semi-supervised (coverage-corrected)	Σ^{-1} noise-aware fine-tune	Any nonlinear $\mathcal{F}$ above	Inherits $\mathcal{F}$'s assumption	Repairs $P_{\text{train}} \to P_{\text{target}}$	Basin trapping; residual sim mismatch	Staged vs sup. and unsup. only; degradation sweep

The latent code is a coefficient matrix

$$C_i = U_c^\top Z_i U_t \in \mathbb{R}^{k_c \times k_t}. \tag{27}$$

The decoder reconstructs

$$\hat{Z}_i = U_c C_i U_t^\top = U_c U_c^\top Z_i U_t U_t^\top. \tag{28}$$

Thus NFPA is a tied linear autoencoder whose projection operator is separable: one projection in channel space and one projection in temporal space.

Definition 4.1 (NFPA model class)**.** *The rank (k_c, k_t) NFPA model class is the set of reconstructions*

$$\mathcal{K}_{k_c,k_t} = \left\{ Z \mapsto U_c U_c^\top Z U_t U_t^\top : U_c^\top U_c = I_{k_c},\ U_t^\top U_t = I_{k_t} \right\}. \tag{29}$$

Equivalently, after vectorization, the reconstruction is the orthogonal projector onto

$$\mathrm{range}(U_c) \otimes \mathrm{range}(U_t) \subseteq \mathbb{R}^C \otimes \mathbb{R}^T. \tag{30}$$

4.2 Covariance and whitening conventions

A trace is a matrix $X_i \in \mathbb{R}^{C \times T}$, but the Gaussian likelihood is defined on the vectorized observation $x_i = \mathrm{vec}(X_i) \in \mathbb{R}^{CT}$, so the noise covariance is in general the full operator $\Sigma = \mathrm{Cov}(\mathrm{vec}(N_i)) \in \mathbb{R}^{CT \times CT}$ and the Mahalanobis loss uses Σ^{-1} on $\mathbb{R}^{CT}$. Three cases must be distinguished, because they are *not* interchangeable:

1. *Full covariance.* Σ is a dense $CT \times CT$ operator capturing arbitrary channel-time noise correlation. Whitening is $z_i = \Sigma^{-1/2} x_i$ in vectorized space, and the NFPA projector acts on z_i. This is the general theory.

2. *Separable (Kronecker) noise.* $\Sigma = \Sigma_c \otimes \Sigma_t$, so $\Sigma^{-1/2} = \Sigma_c^{-1/2} \otimes \Sigma_t^{-1/2}$ and whitening factorizes; in matrix form $Z_i = \Sigma_c^{-1/2} X_i \Sigma_t^{-1/2}$. Here whitening and the NFPA channel/time projectors commute.

3. *Block-diagonal temporal PSD (current implementation).* Noise is independent across channels and stationary in time, so $\Sigma = I_C \otimes \Sigma_t$ with Σ_t the temporal PSD covariance, and whitening is applied in frequency space along time, per channel: $Z_i = X_i \Sigma_t^{-1/2}$.

The restricted bridge theorem below is stated for general Σ (case 1); the controlled experiment of Section 5 uses case 3, which is an *implementation choice* — it assumes no cross-channel noise correlation — not a restriction of the theory. In a full implementation with correlated channels, the channel noise correlations are handled by whitening with the full $\Sigma^{-1/2}$ (case 1) or its Kronecker factors (case 2) *before* applying the separable signal projector; the separability prior on U_c, U_t is a statement about the *signal* subspace and is independent of whether the *noise* covariance is separable. When the noise is non-separable, whitening does not commute with the channel/time projectors, and the ALS updates act on the whitened tensor $z_i = \Sigma^{-1/2} x_i$ in vectorized space.

4.3 Objective

NFPA is trained in whitened space with Euclidean reconstruction loss:

$$\mathcal{L}_{\mathrm{NFPA}}(U_c, U_t) = \frac{1}{N} \sum_{i=1}^{N} \left\| Z_i - U_c U_c^\top Z_i U_t U_t^\top \right\|_F^2 . \tag{31}$$

In native detector coordinates this is the same as the Mahalanobis loss

$$\frac{1}{N} \sum_i \|X_i - \hat{X}_i\|_{\Sigma^{-1}}^2 . \tag{32}$$

Thus NFPA does not change the likelihood. It changes only the admissible projector class.

Proposition 4.1 (Restricted bridge theorem for NFPA)**.** *Let $Z_i = \Sigma^{-1/2} X_i$ be whitened observations. The minimizers of Eq.* (31) *are exactly the minimizers of the Paper 1 linear-autoencoder/EMPCA objective restricted to the Kronecker-separable subspaces*

$$\mathrm{range}(U_c) \otimes \mathrm{range}(U_t).$$

Consequently, the NFPA optimum equals the EMPCA optimum if and only if at least one rank-$k_c k_t$ EMPCA minimizer has a Kronecker-separable subspace of this form. Otherwise NFPA is a structured approximation to EMPCA; no claim of general equivalence to unrestricted EMPCA is made.

Proof. For any $Z \in \mathbb{R}^{C\times T}$,

$$\operatorname{vec}(U_c U_c^\top Z U_t U_t^\top) = (U_t U_t^\top \otimes U_c U_c^\top)\operatorname{vec}(Z). \tag{33}$$

The Kronecker product above is the orthogonal projector onto $\operatorname{range}(U_t) \otimes \operatorname{range}(U_c)$ with rank $k_t k_c$. Therefore Eq. (31) is exactly the ordinary PCA reconstruction loss in whitened space, but with the projector restricted to the Kronecker subset of all rank-$k_t k_c$ projectors. Paper 1 proves that the unrestricted whitened projector problem is EMPCA in native coordinates. Restricting the projector class gives the stated result. □

Remark 4.1 (Subspace mismatch is the cost of the prior, not of the metric)**.** *When the true EMPCA subspace is not Kronecker-separable, NFPA is a structured approximation. The resulting subspace mismatch is not a failure of the noise metric; it measures the cost of the architectural prior. This distinction is the reason the experiments in Section 5 report both likelihood-aligned residuals (which test the metric) and principal angles to EMPCA (which test the prior).*

Remark 4.2 (Why this is Paper 2 material)**.** *The exact bridge theorem in Paper 1 concerns the full Grassmannian* $\operatorname{Gr}(k, d)$*. NFPA replaces this by a product geometry,* $\operatorname{Gr}(k_c, C) \times \operatorname{Gr}(k_t, T)$*, together with the tensor-product embedding into* $\operatorname{Gr}(k_c k_t, CT)$*. This is an architectural prior: channel and temporal structure are assumed to factor. That is why NFPA is the first structured extension rather than part of the core Paper 1 theorem.*

4.4 Alternating least-squares training

The implementation trains Eq. (31) by alternating least squares. For fixed U_c, define

$$D_i = U_c^\top Z_i \in \mathbb{R}^{k_c\times T}. \tag{34}$$

Updating U_t reduces to

$$\max_{U_t^\top U_t = I_{k_t}} \sum_{i=1}^{N} \|D_i U_t\|_F^2. \tag{35}$$

Stacking the D_i matrices into an $(N k_c) \times T$ matrix gives a standard Rayleigh-Ritz problem; U_t is given by the top k_t right singular vectors of the stack.

For fixed U_t, define

$$E_i = Z_i U_t \in \mathbb{R}^{C \times k_t}. \tag{36}$$

Updating U_c reduces to

$$\max_{U_c^\top U_c = I_{k_c}} \sum_{i=1}^{N} \|U_c^\top E_i\|_F^2, \tag{37}$$

so U_c is given by the top k_c right singular vectors of the stacked $(Nk_t) \times C$ matrix. Each subproblem is solved exactly, so the training objective is nonincreasing. As usual for alternating projections on a nonconvex product manifold, this is a local optimizer, not a guarantee of the global Kronecker optimum.

4.5 Rank-one NFPA and the optimal filter

Theorem 4.2 (Rank-one NFPA recovers OF under separability)**.** *Assume the whitened signal template is separable,*

$$\tilde{s} = v\, t^\top, \qquad v \in \mathbb{R}^C, \quad t \in \mathbb{R}^T, \tag{38}$$

and the rank-one signal model is

$$Z_i = A_i \tilde{s} + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, I). \tag{39}$$

In the population, rank-one NFPA with $k_c = k_t = 1$ learns

$$u_c = v/\|v\|, \qquad u_t = t/\|t\|$$

up to independent signs. Its latent coefficient is proportional to the optimal-filter amplitude estimator.

Proof. The rank-one NFPA basis vector in vectorized whitened space is $u_t \otimes u_c$. The population signal covariance has top eigendirection $\mathrm{vec}(\tilde{s})/\|\tilde{s}\| = (t \otimes v)/(\|t\|\|v\|)$, which is itself Kronecker-separable. Therefore the Kronecker restriction does not exclude the rank-one maximum-likelihood direction. Projection onto this direction gives the matched-filter coefficient in whitened space, which is the optimal-filter amplitude estimator up to the chosen normalization. □

4.6 Gauge freedom

NFPA has a product gauge symmetry. For any orthogonal matrices $Q_c \in O(k_c)$ and $Q_t \in O(k_t)$,

$$U_c \mapsto U_c Q_c, \qquad U_t \mapsto U_t Q_t \tag{40}$$

leaves the projectors $U_c U_c^\top$ and $U_t U_t^\top$ unchanged. The code matrix transforms as

$$C_i \mapsto Q_c^\top C_i Q_t, \tag{41}$$

but the reconstruction is invariant. This is the Kronecker analogue of the global $U(k)$ gauge freedom of EMPCA, with the important caveat that NFPA has a smaller product gauge group rather than arbitrary rotations of the full $k_c k_t$-dimensional latent space.

4.7 Approximation error versus metric error

The discrepancy between NFPA and EMPCA observed in experiments decomposes into conceptually distinct contributions, and conflating them leads to wrong conclusions:

1. *Metric mismatch*: training with isotropic MSE instead of the Mahalanobis loss. This is the Paper 1 failure mode and is entirely avoidable; NFPA trained in whitened space has none of it.

2. *Architecture approximation error*: restricting the projector class to Kronecker-separable subspaces. This is the deliberate price of the factorization prior.

3. *Statistical error*: finite training samples.

4. *Optimization error*: alternating least squares converges to a local optimum on a nonconvex product manifold.

Schematically, for the learned reconstruction family $\hat{\mathcal{M}}$ relative to the target $\mathcal{M}$,

$$\operatorname{dist}(\hat{\mathcal{M}}, \mathcal{M}) \lesssim \epsilon_{\text{approx}} + \epsilon_{\text{stat}} + \epsilon_{\text{opt}}, \tag{42}$$

once the metric is correct; using the wrong metric adds a bias term that does not vanish with more data. For the linear subspaces of this section, the target quantitative statement is

$$\sin \Theta(\mathcal{S}_{\text{NFPA}}, \mathcal{S}_{\text{EMPCA}}) \leq F(\epsilon_{\text{kron}}, \Delta_\lambda), \tag{43}$$

where ϵ_{kron} is the best Kronecker (Tucker-2) approximation error to the EMPCA projector and Δ_λ is the relevant spectral gap. We state (43) as a proposed stability interpretation, in the spirit of the Davis–Kahan deformation-stability bound of Paper 1; a proof, plausibly combining a Tucker approximation bound (Kolda and Bader, 2009) with Davis–Kahan perturbation, is left to future work and is *not* claimed here.

5 NFPA Experiments

The completed experiments are organized around six questions:

A. Does NFPA implement the restricted bridge objective of Proposition 4.1?

B. Does rank-one NFPA recover OF when the whitened template is separable (Theorem 4.2)?

C. Does metric reversal persist under the same Kronecker architecture when only the training metric changes?

D. How much of the EMPCA subspace is captured by NFPA, and how much discrepancy is due to the Kronecker restriction (Remark 4.1)?

E. Are the learned channel and time bases physically interpretable (as projectors, per Section 3.4)?

F. Does NFPA improve likelihood-aligned metrics while MSE-trained factored models win only raw MSE?

Questions A, B, and E are addressed by the consistency panels in Figure 1; C and F by the metric-reversal comparison in Table 3; and D by the principal-angle analysis in Table 4. The required NFPA sweeps are specified in Section 5.5; cross-architecture and coverage experiments are deferred to Section 8.

5.1 Setup

The current notebook uses a controlled two-channel problem:

$$C = 2, \qquad T = 256, \qquad f_s = 1\,\text{kHz}.$$

The signal template is an asymmetric pulse. Its channel weights are $v = (1.0, 0.65)$, and event amplitude is $A \sim \mathcal{N}(1, 0.3^2)$. Noise is Brownian with $S_n(f) \propto 1/f^2$, and whitening is applied by the matched frequency-domain filter. The dataset has 800 training events and 300 test events. The NFPA model uses $k_c = 2$, $k_t = 3$, and therefore $k = k_c k_t = 6$ latent dimensions.

The baselines are:

- EMPCA: PCA on whitened data, the unrestricted likelihood-aligned linear subspace.
- NFPA: Kronecker-constrained ALS on whitened data.
- IsoAE: the same Kronecker architecture trained on raw data with MSE.
- IsoPCA: PCA on raw data.

5.2 Metric reversal (Questions C and F)

Table 3: Held-out reconstruction metrics from the NFPA notebook (single seed, one train/test split). The likelihood-aligned metric is the whitened residual χ^2; raw MSE is the misspecified isotropic metric under colored noise. Raw MSE is shown in scientific notation; the isotropically trained models fall below the previous five-decimal display precision ($<10^{-5}$), and exact values should be re-exported. The whitened-χ^2 margins are sub-percent and are not yet established as significant (see text).

Method	Whitened residual χ^2	Raw MSE
EMPCA	0.9929	4.95×10^{-3}
NFPA	0.9916	4.95×10^{-3}
IsoAE	0.9995	$<10^{-5}$
IsoPCA	1.0007	$<10^{-5}$

The robust, reproducible effect is the *disagreement in ranking*: the raw-MSE-optimal models (IsoAE, IsoPCA, at $< 10^{-5}$ raw MSE) are not the whitened-χ^2-optimal models, and the two orderings invert. This is the Paper 1 phenomenon — isotropic MSE rewards reconstruction in high-variance noise directions while the likelihood metric rewards high-information directions —

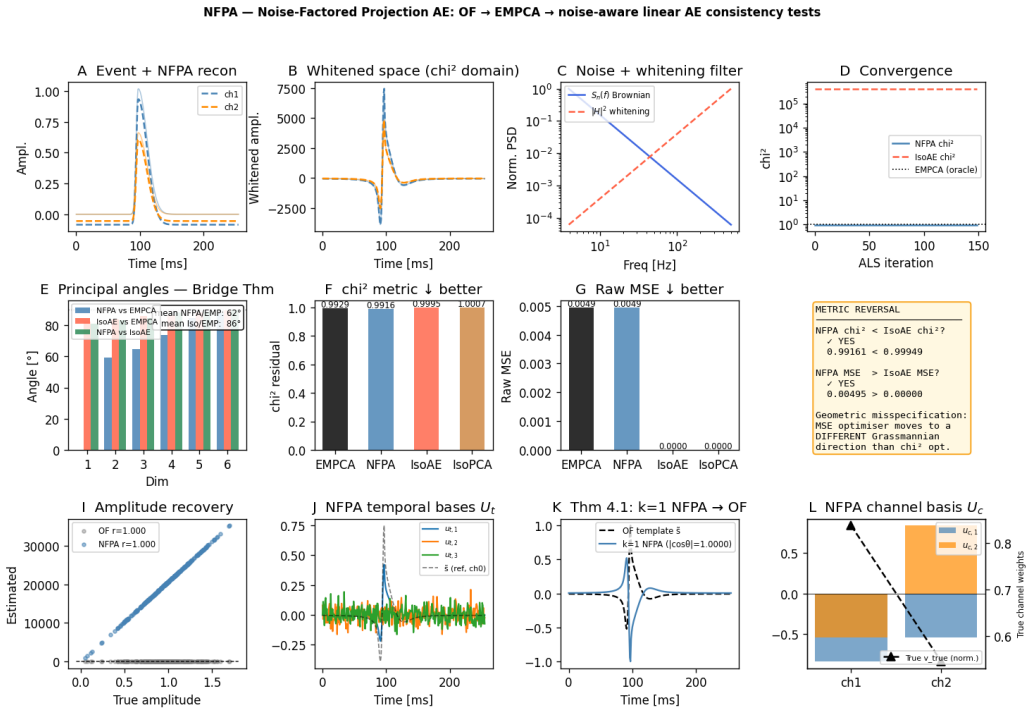


Figure 1: NFPA consistency tests from the current notebook (single seed). Panels A–D show reconstruction, whitening, and ALS convergence; panels E–H compare subspace angles and metric reversal; panels I–L show amplitude recovery, learned bases, and the rank-one NFPA/OF template match. Any panel labeled "Bridge Thm" denotes the *Restricted NFPA bridge* of Proposition 4.1, not the full EMPCA bridge theorem of Paper 1. This dense multi-panel figure is a working diagnostic; for submission it should be split into three publication figures — metric reversal, principal-angle/subspace comparison, and rank-one NFPA/OF recovery — with the full panel grid retained only in an appendix (Section 8).

and it persists when the architecture is held fixed and only the training metric changes. We are deliberately cautious about *magnitudes*: the EMPCA–NFPA whitened-χ^2 difference (0.9929 vs 0.9916, about 0.1%) is a single-seed point estimate within plausible seed-to-seed scatter and should not be read as a significant ordering until reported as a mean with a confidence interval over seeds (Section 8). The effect size that matters here is the order-of-magnitude raw-MSE gap combined with the inverted whitened-χ^2 ranking, not the third-decimal differences.

5.3 Subspace alignment (Question D)

Table 4: Principal angles in whitened space. NFPA captures the dominant EMPCA direction exactly in this experiment, but the remaining angles are moderate to large because the Kronecker factorization is a constrained approximation to the full EMPCA subspace.

Comparison	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	Mean
NFPA vs EMPCA	0.0	59.3	64.6	73.8	84.8	87.8	61.7
IsoAE vs EMPCA	82.1	83.7	86.0	86.6	88.2	89.4	86.0
NFPA vs IsoAE	80.9	83.1	84.4	88.2	88.7	89.9	85.9

These numbers are the most important reason to avoid overclaiming. NFPA does not recover the entire EMPCA subspace in this experiment. Rather, it recovers the dominant signal direction and remains much closer to EMPCA than the MSE-trained factored model. The mean angle of 61.7° is not "small" in an absolute full-subspace sense; it is evidence that the Kronecker constraint is active. The Paper 2 theorem should therefore include an approximation bound, for example in terms of the best Kronecker approximation error to the EMPCA projector. That bound remains open in this draft.

5.4 Rank-one and amplitude checks (Questions A, B, and E)

The rank-one experiment gives the cleanest exact numerical check:

$$|\cos\theta(\mathrm{NFPA}_1, \mathrm{OF})| = 1.0000. \tag{44}$$

The amplitude correlation is also 1.0000 for both OF and rank-one NFPA. This supports Theorem 4.2 in the controlled separable setting. The higher-rank NFPA latent code also contains an amplitude coordinate with correlation $r = 1.0000$ in the current high-SNR setup.

Remark 5.1 (Current experimental limitations)**.** *The present NFPA experiment is useful but not submission-complete. It uses one synthetic high-SNR setting, one train/test split, and one small channel count. The SNR is approximately* 914*, so amplitude recovery is nearly trivial. A submission-ready Paper 2 should add a sweep over SNR, noise color, channel count, rank pair* (k_c, k_t)*, and multiple seeds. The present result is best treated as a proof-of-concept and as a guide for the theorem statement.*

5.5 Required NFPA experiments

The present result is a proof-of-concept on one synthetic, high-SNR, two-channel, single-seed setting, and it leaves the central NFPA claims only partially tested. Before submission the following NFPA-specific sweeps are required; each is tied to a specific claim, and all should report mean values with bootstrap or across-seed confidence intervals over at least ~ 20 seeds and multiple train/test splits. (Cross-architecture and coverage experiments are deferred to Section 8.)

1. *SNR sweep* (low, medium, high; the current SNR ≈ 914 makes amplitude recovery trivial). Tests whether the rank-one NFPA/OF amplitude correspondence and the metric reversal survive when amplitude recovery is nontrivial. Metrics: amplitude RMSE and correlation, whitened χ^2, raw MSE.

2. *Noise-color sweep*: white, pink $(1/f)$, Brownian $(1/f^2)$, and a measured or realistic detector-like PSD. Tests that metric reversal appears under colored/correlated noise and vanishes for white noise.

3. *Channel-count sweep*: $C = 2, 4, 8, 16$. Tests whether the channel factorization scales and helps as the channel dimension grows.

4. *Rank-pair sweep*: vary (k_c, k_t) and compare total rank $k = k_c k_t$ against unrestricted EMPCA at the same k. Separates the rank effect from the factorization effect and quantifies the cost of the Kronecker prior at matched rank.

5. *Nonseparable-template test*: construct a signal family whose EMPCA subspace is *not* Kronecker-separable (e.g. $s = vt^\top + \alpha R$ with R off the separable component), and sweep α. Makes the cost of the NFPA prior visible and tests that the rank-one NFPA/OF match degrades gracefully as separability fails.

6. *Seeds and splits*: repeat all of the above over multiple seeds and splits, reporting confidence intervals and effect sizes rather than point rankings.

Reporting for every cell should list N_{train}, N_{test}, C, T, the noise model and SNR definition, the whitening method, the rank (k_c, k_t) and equivalent k, the number of seeds, and the confidence-interval type.

6 Nonlinear Temporal Denoisers: ResNet and U-Net Priors

NFPA is the structured *linear* member of the hierarchy (24): it imposes a Kronecker channel-time product prior but no nonlinearity, and it does not exploit temporal locality. The next step in the hierarchy relaxes linearity while adding the temporal-grid prior of Section 3.1: nonlinear temporal denoisers built from residual (ResNet) and multiscale encoder–decoder (U-Net) blocks. In the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}} \to P_{\text{target}})$ these models keep the residual geometry — they must still be trained under Σ^{-1}, by whitening the input or by using the Mahalanobis loss — and keep the temporal-grid domain assumption, but enlarge $\mathcal{F}$ from a separable linear projection to a nonlinear local map. They are the natural detector denoisers *between* NFPA and a fully global Transformer, and the case study of Section 7 is a concrete U-Net instance.

6.1 ResNet: residual corrections to a baseline

A residual block computes

$$x \;\mapsto\; x + r_\theta(x), \tag{45}$$

and a ResNet denoiser is most naturally read as learning a *correction* to a baseline reconstruction rather than a reconstruction from scratch. Initializing or anchoring the baseline at the linear noise-aware estimate — the OF/EMPCA or NFPA output — makes this explicit: the network learns only

the residual structure the linear model misses, such as amplitude-dependent pulse shape or mild nonlinearity, while the identity skip keeps the linear baseline recoverable. Its architecture prior is therefore temporal locality together with a near-identity inductive bias, which is well matched to the high-SNR detector regime where the linear baseline is already close. The characteristic failure mode is structural: if convolution is applied across the *channel* axis without a valid channel grid or graph, it imposes a false locality prior on channels that are not grid neighbors (Section 3.2). Temporal convolution is justified by the time grid; channel convolution is justified only when the channel axis genuinely has grid or graph structure.

6.2 U-Net: multiscale temporal detail with skip connections

A U-Net is a convolutional encoder–decoder in which downsampling captures slow baseline and decay structure while skip connections carry full-resolution detail across the bottleneck. The skip connections are the key prior: a plain bottleneck autoencoder compresses the trace through a narrow code and discards sharp, local pulse detail together with the noise, whereas the U-Net skips preserve the high-frequency signal content needed for an accurate amplitude or timing estimate. This is the bottleneck-versus-skip distinction recorded in Table 2 and observed directly in the case study, where a bottleneck autoencoder was rejected in favor of a U-Net. As with ResNet, the multiscale temporal convolutions are justified on the time axis only; the channel axis should be left fully connected, or handled with an explicit graph operator, unless it has genuine adjacency.

6.3 Why these are not magic

Both architectures inherit the same two requirements as every other row of the framework. First, they must be trained in the correct noise geometry: an isotropic MSE loss reintroduces the Paper 1 metric reversal regardless of depth or skip connections, so the input is whitened or the loss is Σ^{-1}-weighted. Second, they inherit the coverage requirement of Section 2.6: a ResNet or U-Net generalizes over shifts, deformations, energies, and positions only to the extent that these are varied in training. ResNet and U-Net are more expressive choices of $\mathcal{F}$, not substitutes for the metric or for coverage; the experiments that would test them are specified in Section 8.

7 Coverage-Corrected Training: A Xenon-TPC Case Study

Sections 4–6 fixed the architecture rows of the framework. This section internalizes the third ingredient, the $P_{\text{train}} \to P_{\text{target}}$ arrow, as a concrete training protocol. We extract it from recent arXiv/NPML work on supervised, semi-supervised, and unsupervised denoising for a single-phase liquid-xenon time projection chamber (TPC) aimed at neutrinoless double-beta decay ($0\nu\beta\beta$) of ^{136}Xe (Parker et al., 2026). We read that work not as an external benchmark but as *a concrete instance of a general coverage-corrected training protocol* that the framework itself generates: it *illustrates*, and is *consistent with*, the separation between likelihood geometry, architecture, and coverage. It does not validate NFPA or any theorem here. All data in the study are *simulated*: what plays the role of target-domain measurements is a set of noisy simulated traces, not data from a constructed TPC, so we describe it as a simulated experimentalist setting.

7.1 The general protocol

Abstracted from the architecture, the protocol is two stages acting on the $P_{\text{train}} \to P_{\text{target}}$ arrow for any nonlinear denoiser $\mathcal{F}$:

- *Stage I (coverage).* Supervised pretraining on an *approximate* simulation. This supplies coarse signal-manifold coverage and places the weights in a good loss basin, but can only identify directions the approximate simulation actually varies (Section 2.6).
- *Stage II (adaptation).* Noise-aware unsupervised or self-supervised fine-tuning on target-domain noisy data, using a covariance-aware objective such as colored MC-SURE (Eq. (47)). This adapts the model to mismatched detector response or missing signal detail *without labels*, repairing the residual $P_{\text{train}} \to P_{\text{target}}$ mismatch to the extent the target-domain objective can.

The protocol is architecture-agnostic: it applies to the U-Net, ResNet, or Transformer rows of Table 2, and the U-Net realization below is one instance. It does not bypass coverage — Stage II cannot recover a physical direction that is absent from *both* the approximate simulation and the target-domain

objective — but it is the framework's practical answer to the common case in which the simulation is only approximately correct.

7.2 Setting

The reconstruction target is the total drift charge of an event, because the fractional charge resolution σ_Q/Q propagates directly into the fractional energy resolution σ_E/E that discriminates the $0\nu\beta\beta$ line from the $2\nu\beta\beta$ continuum and backgrounds (Parker et al., 2026). Each event is read out on a single 32-electrode tile sampled at 1377 time steps, so a trace is an element of the product domain

$$\Omega = V_{\text{channel}} \times V_{\text{time}}, \qquad |V_{\text{channel}}| = 32, \quad |V_{\text{time}}| = 1377, \tag{46}$$

of dimension $32 \times 1377 = 44{,}064$ — the same channel-time product domain that NFPA factorizes (Section 3.2). The instrumentation noise is colored, with a $1/f$ power spectral density estimated from prototype electronics, and is further shaped by 12-bit quantization; it therefore has a nondiagonal covariance Σ in the sense of Paper 1, not white noise. A distinctive feature is the very high signal-to-noise ratio (pre-denoising PSNR $\approx 50\,\text{dB}$): a typical event deposits $\sim 1.2 \times 10^5$ electrons against a per-sample noise scale of order 150 electrons. To avoid biasing the denoiser toward the physical Q-value, training events span a uniform energy range (700–4000 keV) while evaluation uses a held-out ensemble at the $0\nu\beta\beta$ Q-value of 2458 keV — an explicit, deliberately broad coverage design over the energy latent.

7.3 The protocol instantiated: a U-Net denoiser

The study combines four components that map one-to-one onto the tuple of this paper; Table 5 summarizes the correspondence.

Likelihood geometry (the metric). The linear baseline is the Wiener / optimal filter: a PSD-weighted, inverse-noise-weighted estimator of a known signal amplitude, identical in form to the rank-one Σ^{-1}-projection of Paper 1. Because it assumes the signal *shape* is known per event and only the amplitude is uncertain, it is not implementable as a per-event denoiser, but it provides the information-theoretic bound the learned models are measured against. The noise geometry re-enters the learned models explicitly through

the unsupervised objective. Stein's unbiased risk estimator (SURE) is the natural noise-aware loss when no clean target is available (Stein, 1981), but its standard form assumes white noise; for the colored detector covariance Σ the study uses the covariance-weighted form

$$\mathcal{L}_{\mathrm{SURE}}(f) = \mathrm{MSE}(f(y), y) - n\ \mathrm{tr}[\Sigma] + 2\ \mathrm{tr}\left[\Sigma\, \frac{\partial f}{\partial y}(y)\right], \tag{47}$$

with the divergence term *colored by the noise covariance*, estimated by a Monte-Carlo (Hutchinson) trace with noise probes drawn from the colored distribution (MC-SURE (Ramani et al., 2008)). Equation (47) is the same statement this paper makes about loss design: the correct objective is dictated by Σ, and replacing Σ by $\sigma^2 I$ optimizes the wrong residual geometry.

Architecture (the admissible map class). The denoiser is a U-Net autoencoder — a convolutional encoder–decoder with skip connections (Ronneberger et al., 2015). Two architectural choices in the study are direct illustrations of the taxonomy in Section 3.5. First, a plain bottleneck autoencoder was tried and rejected: compressing through a small bottleneck discarded signal structure together with noise, so the skip connections of the U-Net — which pass full-resolution detail across the bottleneck — were required to preserve sharp pulse features. This is exactly the CNN/U-Net failure mode in Table 2 (a too-aggressive bottleneck), resolved by the skip connections that preserve full-resolution detail. Second, convolution is applied as a *one-dimensional* operation along time only, while the channel axis is left fully connected rather than treated with 2D convolution, because the electrodes "do not have a pixel-like adjacency relationship." In the language of Section 3.2, this is a deliberate refusal to impose a grid prior on a non-grid domain: the temporal axis carries locality and shift structure, but the channel axis is a graph whose geometry is set by charge induction and covariance, not by sensor index. The architecture thus encodes precisely the domain priors the detector supports, and no more.

Coverage (which variation is identifiable). Supervised training needs clean–noisy pairs, which a built detector cannot provide. The study addresses this with a two-phase semi-supervised scheme: Phase I pretrains the U-Net in supervised mode on an *approximate* simulation, and Phase II continues training with the unsupervised colored-SURE objective of Eq. (47) on target-domain noisy measurements whose clean signals are never observed. This is

the achievability-versus- selection mechanism of Section 2.6 made operational: the approximate simulation supplies coverage (it places the model in the right loss basin), while the unsupervised target-domain stage corrects the residual simulation–target mismatch without ever requiring labels. To probe sensitivity to coverage quality, the approximate simulation is degraded by a low-pass filter to a controlled information-loss level $D \in \{0.1, 10, 30, 55\}\%$, a synthetic proxy for a simulation that captures coarse but not fine signal structure.

Table 5: The Xenon-TPC denoising study (Parker et al., 2026) as an instance of the coverage-corrected protocol, mapped onto the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}} \to P_{\text{target}})$. It illustrates the separation; it does not test the NFPA theorems.

Ingredient	In this paper	In the Xenon study
Likelihood geometry	Σ^{-1} residual metric; OF as rank-one Σ^{-1} projection	Colored $1/f$ covariance Σ; Wiener/OF bound; covariance-weighted MC-SURE divergence, Eq. (47)
Architecture prior	Kronecker channel-time factorization (NFPA); taxonomy of Sec. 3.5	U-Net with skip connections; 1D temporal convolution; fully-connected (non-grid) channel axis
Coverage condition	Training must excite the latent directions to be reconstructed (Sec. 2.6)	Broad energy sampling; semi-supervised target-domain adaptation under controlled simulation degradation D

7.4 What the results do and do not show

Table 6 reproduces the reported resolutions. Three features are relevant to this paper. First, every learned model sits between the Wiener-filter bound and the conventional filters, confirming that the linear PSD-weighted estimator of Paper 1 is the right reference point even for nonlinear denoisers. Second, performance degrades *monotonically* as the Phase-I simulation is made less accurate (σ_E/E rising from 0.86% at $D = 0.1\%$ to 1.63% at $D = 55\%$): this

is a clean empirical signature of the coverage condition — when the training distribution misrepresents the signal manifold, the achievable reconstruction quality is bounded by how well the target-domain stage can repair the mismatch. Third, the fully unsupervised model, which in principle has enough information in the noisy data to learn the optimal denoiser, underperforms the semi-supervised models in practice because its colored-SURE objective is harder to optimize (it settles in a higher-loss basin). This is a concrete instance of the achievability-versus-selection distinction of Section 2.6: a correct noise-aware objective makes good reconstruction *achievable*, but the architecture, initialization, and optimization still decide which admissible solution is *selected*.

Table 6: Reported charge and energy resolutions for the Xenon-TPC denoisers (Parker et al., 2026). The Wiener filter is a non-implementable bound; "semi-supervised D" denotes the Phase-I simulation information-loss level. All entries are from simulation.

Method	σ_Q/Q	σ_E/E
Wiener (optimal) filter — bound	0.477%	0.680%
Supervised U-Net	0.859%	0.802%
Semi-supervised U-Net, D=0.1%	1.014%	0.863%
Semi-supervised U-Net, D=10%	1.299%	0.989%
Semi-supervised U-Net, D=30%	1.479%	1.075%
Semi-supervised U-Net, D=55%	2.536%	1.629%
Unsupervised U-Net	2.773%	1.761%
Trapezoidal filter	4.5%	2.746%
No denoising	13.8%	8.230%

What the example does *not* establish is equally important. It does not validate NFPA, the Kronecker factorization, or any nonlinear theorem of this paper; the U-Net is not claimed to be optimal, and the colored-SURE objective is demonstrated, not proven, to recover the noise geometry. Because the target domain is itself simulated, the study is a simulated experimentalist setting, not evidence on real detector data. What it does motivate is the *general training strategy* of Section 7.1, which the framework can apply to any nonlinear denoiser — U-Net, ResNet, or Transformer — and which Section 8 turns into a controlled future experiment. In the tuple language, the noise covariance defines the residual geometry, the U-Net defines the admissible

nonlinear maps, and the semi-supervised stages act on the $P_{\text{train}} \to P_{\text{target}}$ arrow — three ingredients that must be specified together.

8 From Framework to Experimental Program

The value of the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}} \to P_{\text{target}})$ is that it generates, rather than merely describes, a concrete experimental program. We do not report these experiments here; the point is that each isolates exactly one ingredient and tests a specific claim. The program varies four axes:

- *loss*: Mahalanobis (Σ^{-1}) versus isotropic MSE — isolates the metric;
- *architecture*: EMPCA / NFPA / ResNet / U-Net / Transformer — isolates inductive bias;
- *coverage*: full versus restricted latent support — isolates identifiability;
- *noise*: matched versus mismatched PSD/covariance between training and evaluation — isolates robustness of the whitening.

Table 7 maps the resulting experiments to the claim each one tests. Across all cells the reported metrics are the held-out whitened residual χ^2, the raw MSE, amplitude and timing bias, reconstruction error versus the held-out latent coordinate, and the full-coverage-minus-restricted generalization gap, with a detector-level energy-resolution or trigger metric where a readout exists.

8.1 The coverage experiment

The coverage condition is the one pillar still supported only by theory (Section 2.6); the following controlled experiment is its minimal test. Generate multi-channel traces from a known generator with explicit latents,

$$s_c(t; A, t_0, p, \gamma) = A\, w_c(p)\, h(t - t_0; \gamma), \qquad n \sim \mathcal{N}(0, \Sigma), \tag{48}$$

where A is amplitude, t_0 arrival time, p position/channel-sharing, and γ a pulse-shape deformation. Train under *restricted support* in one latent at a time — fixed t_0, narrow energy range, fixed position, restricted shape γ, or restricted noise/PSD state — and evaluate both in-support and on a full-coverage test set that varies the held-out latent. Report reconstruction

Table 7: The framework-generated experimental program. Each experiment varies one axis of the tuple and tests one claim. Entries are *hypotheses* to be tested, not results.

Experiment	Axis varied	Expected outcome (hypothesis)	Claim tested
Metric reversal	Loss	MSE wins raw MSE but loses whitened χ^2 under colored noise; vanishes for white noise	Loss geometry is necessary (Paper 1; persists for NFPA)
Coverage violation	Coverage	Good in-support; large gap on a full-coverage test when a latent is unvaried	Coverage condition (Section 2.6)
Architecture ladder	Architecture	Architecture gaps small at full coverage, widening under restriction	Inductive bias selects among admissible maps
Noise mismatch	Noise	Degradation when the whitening Σ mismatches the evaluation noise	Whitening requires a reliable Σ
Coverage-corrected training	$P_{\text{train}} \to P_{\text{target}}$	Staged > supervised-only and unsupervised-only; degrades as the Stage-I simulation is degraded	The $P_{\text{train}} \to P_{\text{target}}$ protocol (Section 7)

error *versus the held-out latent coordinate*, and compare Mahalanobis-trained against MSE-trained versions of the same architecture (flat AE, NFPA, U-Net). The predicted result is the central distinction of the paper: the correct Mahalanobis geometry improves in-support reconstruction but does *not* repair the missing latent direction — error grows along the unvaried coordinate regardless of loss — with the worst case being restricted coverage *and* isotropic MSE, where both failures stack.

8.2 The coverage-corrected training experiment

The protocol of Section 7.1 becomes a controlled experiment by following the Xenon-TPC recipe on synthetic data with a *known* mismatch. Stage I: train a U-Net or ResNet supervised on an approximate simulation degraded by a controlled amount (e.g. a low-pass filter of tunable strength, as in the case study). Stage II: fine-tune with a covariance-aware unsupervised objective — colored MC-SURE, Eq. (47) — on target-domain noisy traces with no labels. Compare against supervised-only training on the degraded simulation and against unsupervised-only training from scratch, and sweep the Stage-I degradation to trace how simulation–target mismatch propagates to the final reconstruction. Metrics: covariance-weighted residual, raw MSE, amplitude and timing bias, and a detector-level energy or trigger metric where available. The hypothesis, consistent with the case study, is that the staged protocol recovers most of the supervised performance and degrades gracefully with simulation mismatch, while unsupervised-only training underperforms because its objective is harder to optimize.

8.3 Reconstructability-aware detector design metrics

The same experimental program can be used for detector design, not only for calibration or denoising. Once the likelihood geometry and architecture prior are fixed, a detector configuration can be evaluated by how its covariance, sampling, channel layout, and readout choices change the distinguishability of physical event directions. For a signal model $s(z)$, the local design metric is the Fisher/pullback matrix already used in Section 3.3,

$$g_z = J_s(z)^\dagger \Sigma^{-1} J_s(z).$$

Changing the PSD, sampling frequency, trace length, channel geometry, or readout noise changes either $J_s(z)$, Σ^{-1}, or both. The design question is there-

fore whether the modified detector increases Fisher rank, separates previously degenerate event directions, reduces the likelihood-aligned reconstruction residual, and improves robustness under realistic coverage shifts.

Table 8: Detector-design quantities induced by the same likelihood-geometry framework. These metrics compare detector configurations by reconstructability rather than by raw denoising error alone.

Design factor	Framework metric
PSD shape	Inverse-PSD weighted distinguishability and threshold
Sampling frequency	Timing-mode Fisher information, aliasing, and phase-mode resolution
Trace length	Baseline, low-frequency drift, and artifact identifiability
Channel layout	Position Fisher information, channel-mode separability, and graph equivariance error
Channel correlations	Precision-graph structure, conditioning, and robustness under channel dropout
Readout noise	Energy resolution, mode degeneracy, and likelihood-aligned residual

In practice this turns a reconstruction model into a design diagnostic only if the model is interpretable enough to say which modes are being used. The NFPA subspace angles, VAE pullback metric, and Transformer local-linear diagnostics therefore play complementary roles: they identify whether design improvements act through known linear likelihood modes, nonlinear event-manifold directions, or learned channel/time attention modes.

8.4 Further diagnostics and figures

The equivariance/invariance, graph-construction, positional-encoding, and VAE pullback-geodesic diagnostics test the appendix material and are listed with the corresponding extensions (Appendices A, B, C). For presentation, the single dense diagnostic figure of Section 5 should be replaced before submission by three publication figures — (i) metric reversal, (ii) principal-angle/subspace comparison, (iii) rank-one NFPA/OF recovery — each with across-seed uncertainty bands, with the full panel grid kept only as an appendix figure.

9 Related Work

We position NFPA against four nearby lines of work; the aim is placement, not a survey.

Covariance-weighted subspace models. Probabilistic PCA (Tipping and Bishop, 1999) and expectation–maximization PCA for heteroscedastic and missing data (Bailey, 2012) generalize PCA to non-isotropic noise, and Paper 1 (Wong, 2026) unifies optimal filtering, EMPCA, and tied noise-aware linear autoencoders as Σ^{-1}-orthogonal projections. NFPA inherits exactly this likelihood metric; its only departure from Paper 1 is the admissible subspace class.

Separable and tensor subspace models. Restricting a subspace to a Kronecker/product structure is a classical idea: Tucker and multilinear factor analysis (Tucker, 1966; Kolda and Bader, 2009), multilinear PCA of tensor objects (Lu et al., 2008), two-dimensional PCA (Yang et al., 2004), and generalized low-rank matrix approximations (Ye, 2005) all learn separable row/column (here channel/time) bases. These methods optimize an unweighted Frobenius reconstruction, which is the Gaussian likelihood only for white noise. NFPA differs in two ways: it optimizes the *whitened* (Mahalanobis) objective, so the factorization is performed in the correct noise geometry, and we give a restricted bridge theorem (Proposition 4.1) tying the product-Grassmannian optimum to the Paper 1 linear-autoencoder/EMPCA optimum. NFPA is best read as a whitened, likelihood-aware Tucker-2 subspace model rather than a new tensor decomposition.

Noise-aware and unsupervised denoising. Stein's unbiased risk estimator (Stein, 1981) and its Monte-Carlo form (Ramani et al., 2008) enable denoiser training without clean targets, and the U-Net (Ronneberger et al., 2015) is the standard encoder–decoder for signal denoising. The Xenon-TPC study of Section 7 (Parker et al., 2026) combines a colored-noise MC-SURE objective with a U-Net and a semi-supervised stage; we use it as an external illustration of the geometry/architecture/coverage separation, not as evidence for NFPA.

Geometric deep learning. We borrow the grids, groups, graphs, geodesics, and gauges vocabulary (Bronstein et al., 2021) only to name which detector priors an architecture encodes (Section 3, Appendix A). We do not propose a new geometric-deep-learning method, and we are explicit that prewhitening fixes the residual metric but does not by itself supply locality, equivariance, or graph structure.

Transformer local linearity and steering. Skifstad et al. (2026) show, for language-model activation steering, that Transformer layer dynamics can often be approximated locally by linear time-varying models and probed with layer-wise Jacobians. Our use is only methodological: we do not import LQR control as a detector algorithm, and we do not treat LLM evidence as detector evidence. Instead, Appendix C.7 adapts the same Jacobian, subspace-alignment, and activation-intervention ideas as diagnostics for whether a detector Transformer uses stable physical modes, noise modes, or artifact-specific nuisance directions.

10 Discussion

In the tuple language of this paper, every method shares the residual geometry Σ^{-1} and differs only in the admissible map class $\mathcal{F}$ and the coverage $P_{\text{train}} \to P_{\text{target}}$ it requires. Along the hierarchy (24), EMPCA uses all rank-k subspaces, NFPA the Kronecker-separable ones, ResNet/U-Net nonlinear temporal-local maps, and a Transformer global attention over whitened tokens, while semi-supervised training acts on the coverage arrow rather than on $\mathcal{F}$. Paper 1 fixed the metric; this paper organizes architecture and coverage around it.

This separation prevents two common overclaims. First, structured models are not automatically equivalent to OF or EMPCA; they inherit the same geometry but optimize over different model classes. Second, a more expressive architecture does not remove the need for the correct likelihood: an isotropic loss reintroduces the Paper 1 metric reversal at any depth, so every row of the hierarchy must still be trained under Σ^{-1}.

The same methodological point appears, *by analogy only*, in recent contrastive-learning theory: Zaliaduonis et al. (Zaliaduonis et al., 2026) argue that objective design alone does not determine representation recovery, because sampling support and inductive bias also matter. The setting differs (a Gaussian detector likelihood rather than InfoNCE, the noise covariance rather

than a latent sphere), so we treat it as a parallel intuition, not a result we inherit; the detector-native statement is the coverage condition of Section 2.6.

The immediate open problem is a quantitative NFPA approximation theorem: a bound of the form

$$\sin\Theta(\mathcal{S}_{\mathrm{NFPA}}, \mathcal{S}_{\mathrm{EMPCA}}) \leq F(\epsilon_{\mathrm{kron}}, \Delta_\lambda),$$

with ϵ_{kron} the best Tucker-2/Kronecker approximation error to the EMPCA projector and Δ_λ the relevant spectral gap, plausibly provable by combining a Tucker approximation bound (Kolda and Bader, 2009) with Davis–Kahan subspace perturbation. We state it as an open problem, not a result; it would turn the empirical principal-angle comparison into a formal restricted bridge theorem. The nonlinear rows — ResNet/U-Net (Section 6) and the Transformer (Appendix C) — are geometry-motivated choices of $\mathcal{F}$, not equivalents of OF or EMPCA; establishing formal nonlinear approximation results is likewise left to future work.

10.1 Limitations

The framework and the completed experiments have explicit limitations.

1. NFPA assumes separable channel-time structure; real detector responses may require nonseparable modes, and the principal-angle results of Table 4 show the Kronecker constraint is active even in the controlled setting.

2. Detector graph symmetries are approximate and may be broken by calibration differences, hardware asymmetry, missing channels, or spatially varying noise.

3. Channel-relabeling equivariance is not the same as physical exchangeability of channels; only the former is assumed anywhere in this paper.

4. The VAE and Transformer proposals require the stated ablations before being claimed as empirical results.

5. Computing the pullback metric g_z requires a differentiable decoder or simulator and may be computationally expensive.

6. Prewhitening assumes a reliable estimate of Σ; covariance mismatch can bias the learned geometry.

7. Non-Gaussian noise and artifacts require likelihoods beyond the Mahalanobis loss; the entire framework is stated under the assumed Gaussian noise model.

10.2 Open theoretical questions

Five theory questions remain open or only partially resolved in this draft.

1. *NFPA approximation bound.* Prove or sharpen the target inequality $\sin\Theta(\mathcal{S}_{\mathrm{NFPA}}, \mathcal{S}_{\mathrm{EMPCA}}) \leq F(\epsilon_{\mathrm{kron}}, \Delta_\lambda)$ by combining Tucker approximation error with spectral-gap stability.

2. *Nonlinear G_{noise} argument.* The Schur-lemma step that selects the χ^2 quadratic form acts on the residual vector, so it does not require the decoder itself to be linear. A fully nonlinear theorem should state this residual-level assumption explicitly.

3. *VAE linear-limit continuity.* Proposition B.2 shows that the reconstruction term converges to the Paper 1 Bridge objective in the deterministic tied-linear limit. A complete VAE statement must also specify the limiting role of the KL regularizer.

4. *Noise-Laplacian PE uniqueness.* It remains open whether there is an analog of Paper 1's symmetry-axiomatic uniqueness theorem for Transformer positional encodings built from Σ^{-1}.

5. *Coverage-corrected achievability.* Section 2.6 gives a first-order detector coverage condition and a local tangent-space argument, but a full achievability-versus-selection theorem still requires both a sharper proof and the coverage experiment of Section 8.

11 Conclusion

We have framed detector reconstruction as a single decision procedure over the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\mathrm{train}} \to P_{\mathrm{target}})$: the likelihood fixes the residual geometry, the architecture fixes the admissible map class, and the training distribution (with any target-domain adaptation) fixes which physical directions are identifiable. The paper makes the following claims and no stronger ones.

First, the correct noise geometry remains *necessary* after architecture is added: an isotropic loss reintroduces the Paper 1 metric reversal regardless of the model class. Second, NFPA is a controlled, interpretable product-domain restriction of the Paper 1 noise-aware linear bridge, replacing the full Grassmannian $\mathrm{Gr}(k, CT)$ by the product geometry $\mathrm{Gr}(k_c, C) \times \mathrm{Gr}(k_t, T)$. Third, this Kronecker restriction changes the admissible subspace class, not the likelihood; equality with EMPCA holds only when an EMPCA minimizer is itself separable. Fourth, in one controlled synthetic experiment, NFPA improves likelihood-aligned reconstruction relative to a same-architecture MSE baseline and captures the dominant EMPCA direction, while the MSE model learns a nearly orthogonal subspace — a proof-of-concept whose required sweeps and confidence intervals are specified, not yet run. Fifth, architecture and coverage jointly determine what nonlinear extensions can represent and identify: ResNet/U-Net, the VAE, the Transformer, and semi-supervised coverage correction are framework-derived directions, each with a stated prior, coverage requirement, failure mode, and minimal experiment, and none is claimed as a validated result here.

The contribution is therefore a framework with one validated Kronecker-factored case study, from which a concrete experimental program follows directly. Future architectures should inherit this likelihood geometry and impose only the domain priors the detector actually supports, under training coverage broad enough to make the targeted physical directions identifiable.

A Extended 5G Taxonomy

This appendix collects the geometric-deep-learning structure deferred from the compressed treatment in Section 3: the group actions, the consolidated detector symmetry map, and the equivariance/invariance diagnostics. None of it is required for the NFPA results of the main text; it is supporting framework for the future directions of Appendices B and C. Throughout, the residual metric Σ^{-1} is held fixed; these structures constrain the hypothesis class $\mathcal{F}$, not the likelihood.

A.1 Groups: time translation and detector point symmetries

At least two group actions are relevant. First, the time-translation group acts by

$$(g_\tau x)(t) = x(t - \tau), \tag{49}$$

continuously ($\mathbb{R}$) or cyclically ($\mathbb{Z}_T$) on the sampled grid. Second, detector rotations and reflections may be approximate symmetries for ring or cylindrical sensor layouts. For a discrete ring of n sensors the natural candidate is the dihedral group D_n of rotations and reflections of the layout, not necessarily continuous $SO(2)$. We stress that these are *candidate* symmetries: hardware asymmetry, channel-dependent calibration, boundary conditions, and spatially varying noise correlations may all break them, and the degree of breaking is an experimental question for any given detector.

When the symmetries hold (exactly or approximately), the desired behaviors are:

- waveform reconstruction: *equivariant* under time shift;
- energy: *invariant* under time shift and detector rotation/reflection;
- t_0: *equivariant* under time shift (by addition);
- position (r, ϕ): *equivariant* under detector rotations/reflections;
- event-class labels (e.g. ER/NR): likely *invariant* under time shift and graph relabeling.

A.2 Detector prior and symmetry map

Table 9 summarizes the detector priors discussed in Section 3 and above, their mathematical form, the desired model behavior, and the architectural implication. The entries are priors of varying status: the noise-covariance row is exact under the assumed Gaussian model, the time-shift row is approximate up to windowing, and the rotation row holds only to the extent that the hardware is symmetric.

Table 9: Detector priors, their transformations, desired model behavior, and architectural implications. Full channel permutation is generally *not* a physical symmetry: what is required is equivariance to graph relabeling, in which node labels, adjacency, coordinates, and covariance are reordered consistently.

Detector prior	Transformation / structure	Desired behavior	Architecture implication
Arrival-time shift	$t \mapsto t + \tau$	trace equivariance, energy invariance, t_0 equivariance	shifted OF, temporal CNN, shift pooling
Pulse deformation	$t \mapsto \varphi(t)$, approximate time-warping	stability, not exact invariance	ResNet, LSTM/RNN, deformation augmentation, derivative modes
Channel relabeling	graph node permutation	equivariance to relabeling, *not* physical exchangeability	GNN, graph Transformer
Detector rotation/reflection	D_n, $SO(2)$, or $O(2)$ depending on geometry	position equivariance, energy invariance	group-equivariant GNN/CNN, if the symmetry is valid
Noise covariance	$Q^\dagger \Sigma^{-1} Q = \Sigma^{-1}$	residual norm invariant	Mahalanobis loss, whitening, noise-aware adjoints
Latent basis rotation	$U \mapsto UQ$	reconstruction invariant	gauge-invariant evaluation
Channel-time factorization	$\mathrm{range}(U_c) \otimes \mathrm{range}(U_t)$	separable channel sharing and temporal pulse evolution	NFPA
Event manifold	$s(z)$, $z = (E, t_0, r, \phi, \gamma, \ldots)$	detector-distinguishability metric g_z	nonlinear AE/VAE, pullback metric, geodesic interpolation
Multi-scale pulse physics	fast rise, slow decay, baseline drift	scale separation	ResNet, U-Net, wavelets, multiscale temporal convolution

A.3 Geometry diagnostics beyond reconstruction error

The symmetry map suggests evaluation metrics that go beyond reconstruction error. For a transformation g acting on inputs through a representation ρ and on outputs through ρ', define the relative equivariance error of a model F,

$$\epsilon_{\mathrm{eq}}(g) = \frac{\|F(\rho(g)x) - \rho'(g)F(x)\|}{\|F(x)\| + \epsilon}, \tag{50}$$

and, for an invariant scalar output $h(x)$ such as energy, the invariance error

$$\epsilon_{\mathrm{inv}}(g) = |h(\rho(g)x) - h(x)|. \tag{51}$$

We propose computing these for time shifts, channel-graph relabelings, detector rotations/reflections in simulation, and pulse-deformation / time-warping stress tests. The task-level expectations follow Section A.1: the reconstructed trace and any denoised latent field over channels should be equivariant; energy should be invariant to time shift and detector rotation; t_0 should be equivariant to time shift; (r, ϕ) should be equivariant to detector rotation; and classification labels should be invariant to relabeling and time shift. These diagnostics are gauge-safe in the sense of Section 3.4: they compare model outputs, not latent coordinates.

B Framework-Derived Extension: Noise-Aware VAE

This appendix is a framework-derived extension, not a validated result. Read through the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\mathrm{train}} \to P_{\mathrm{target}})$, a noise-aware VAE is specified by: *architecture prior* — a nonlinear decoder with a latent prior, replacing the linear EMPCA subspace and the NFPA product subspace by the image of a nonlinear map; *detector structure assumed* — a low-dimensional event manifold (Section 3.3); *coverage required* — dense sampling of that manifold, since off-support latents have no training signal; *failure mode* — latent "teleportation" through off-manifold regions under a flat latent geometry; and *minimal experiment* — compare isotropic and noise-aware VAEs on the same controlled pulse data with the pullback-geodesic and coverage tests of Section 8. The residual metric Σ^{-1}, and therefore the pullback geometry of Section 3.3, is unchanged; the extension changes only $\mathcal{F}$.

The VAE extension changes the decoder likelihood, not the KL term. A standard VAE often assumes

$$p_\theta(x \mid z) = \mathcal{N}(D_\theta(z), \sigma^2 I).$$

For detector traces with known covariance, this is the wrong observation model. The noise-aware VAE uses

$$p_\theta(x \mid z) = \mathcal{N}(D_\theta(z), \Sigma).$$

Let $r_\theta(x, z) = x - D_\theta(z)$. The evidence lower bound becomes

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} &= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi \,\|\, p) && (52) \\ &= -\frac{1}{2}\mathbb{E}_{q_\phi}\left[r_\theta^\top \Sigma^{-1} r_\theta\right] - \frac{1}{2}\log|\Sigma| \\ &\quad - \mathrm{KL}(q_\phi \,\|\, p) + \text{const.} && (53) \end{aligned}$$

For fixed Σ, the log determinant is constant during training. The reconstruction term is exactly the Mahalanobis loss. From an information-geometry perspective (Amari and Nagaoka, 2000), the Gaussian family with fixed Σ forms a flat statistical manifold whose Fisher information metric on the mean parameter is Σ^{-1}; the Mahalanobis χ^2 distance is the squared geodesic distance under this metric. The noise-aware ELBO is therefore selected by three compatible arguments: maximum likelihood, G_{noise} symmetry, and information geometry.

Proposition B.1 (Architecture-neutral loss selection)**.** *If the decoder residual loss is quadratic in $r = x - D_\theta(z)$ and invariant under the noise-isometry group*

$$G_{\text{noise}} = \{Q : Q^\top \Sigma^{-1} Q = \Sigma^{-1}\},$$

then the only admissible positive-definite quadratic form is proportional to Σ^{-1}.

Proof. The proof is the same Schur-lemma argument as in Paper 1. The architecture does not enter: the condition is imposed on the residual vector r. Whitening converts G_{noise} into the full orthogonal group; the only quadratic form invariant under all orthogonal transformations is a scalar multiple of the identity. Transforming back gives a scalar multiple of Σ^{-1}. □

Remark B.1 (Pullback metric on latent space)**.** *For a nonlinear decoder $D_\theta : \mathcal{Z} \to \mathcal{X}$, the χ^2 reconstruction loss induces a geometry on latent space through the pullback of Σ^{-1} by the decoder Jacobian, the metric g_z of Eq.* (22)*:*

$$g_z = J_{D_\theta}(z)^\top \Sigma^{-1} J_{D_\theta}(z). \tag{54}$$

This is the nonlinear analog of Paper 1's linear metric geometry, and it links VAE latent geometry, uncertainty, identifiability, and geodesic interpolation to the detector likelihood. It is the noise-induced Riemannian metric on $\mathcal{Z}$: latent directions that map to high-SNR directions in observation space are metrically enlarged. The isotropic VAE case, $\Sigma = \sigma^2 I$, induces the corresponding flat Euclidean pullback and assigns equal metric weight to signal and noise directions. This is the nonlinear version of the metric-reversal mechanism in Paper 1 and matches the pullback-geometry viewpoint used by Wurgaft et al. (Wurgaft et al., 2026) for neural representation steering.

B.1 Geometry diagnostics for nonlinear latent models

The pullback metric suggests concrete diagnostics for nonlinear experiments, not just a different loss. For two latent codes z_a, z_b, the intrinsic noise-aware length of a latent path $\gamma : [0, 1] \to \mathcal{Z}$ is

$$L_\Sigma(\gamma) = \int_0^1 \sqrt{\dot\gamma(t)^\top g_{\gamma(t)} \dot\gamma(t)}\, dt. \tag{55}$$

A Wurgaft-style isometry diagnostic is to estimate the induced geodesic distance $d_{\text{latent}}(z_a, z_b)$, for example by shortest paths on a nearest-neighbor graph whose edge weights approximate L_Σ, and compare it with an observation-space behavior distance such as

$$d_{\mathcal{X}}(z_a, z_b)^2 = \left(D_\theta(z_a) - D_\theta(z_b)\right)^\top \Sigma^{-1} \left(D_\theta(z_a) - D_\theta(z_b)\right). \tag{56}$$

If the model has learned a useful detector manifold, noise-aware training should preserve this geometry more faithfully than isotropic MSE training: pairwise latent geodesic distances should correlate better with likelihood-aligned observation distances, and their distortions should be smaller.

The same metric gives an interpolation and steering test. Given endpoints z_a, z_b, compare the straight latent segment $\gamma_{\text{lin}}(t) = (1 - t)z_a + tz_b$ with a path that minimizes the energy

$$E_\Sigma(\gamma) = \int_0^1 \dot\gamma(t)^\top g_{\gamma(t)} \dot\gamma(t)\, dt. \tag{57}$$

The Wurgaft et al. failure mode is that a flat Euclidean path can "teleport" through regions that are not natural under the model. In detector language this can be measured by a naturalness energy. Let $\mathcal{M} \subset \mathcal{X}(\Omega)$ be a reference signal manifold, either a known pulse family or an empirical nearest-neighbor manifold. Define

$$d_\Sigma(y, \mathcal{M})^2 = \min_{m \in \mathcal{M}} (y - m)^\top \Sigma^{-1} (y - m), \tag{58}$$

and, for a decoded path $y(t) = D_\theta(\gamma(t))$,

$$E_{\mathrm{nat}}(\gamma) = \int_0^1 d_\Sigma(y(t), \mathcal{M})^2 \sqrt{\dot{y}(t)^\top \Sigma^{-1} \dot{y}(t)}\, dt. \tag{59}$$

This turns "off-manifold" behavior into a detector-weighted quantity rather than a visual judgment. A successful pullback-geodesic interpolation should reduce E_{nat} relative to a straight Euclidean latent interpolation while preserving endpoint quality.

Finally, the same setup gives a factored-control diagnostic. Let $h_b(D_\theta(z))$ denote interpretable detector factors such as amplitude, arrival time, pulse width, or channel mode. Perturbing a latent coordinate or subspace a defines a finite-difference leakage matrix

$$L_{ab} = \frac{h_b(D_\theta(z + \epsilon e_a)) - h_b(D_\theta(z - \epsilon e_a))}{2\epsilon}. \tag{60}$$

Good factored representations should have concentrated rows or blocks of L_{ab}: steering one intended factor should not move unrelated factors strongly. For NFPA, the same diagnostic can be applied to channel and temporal modes; for a VAE, it can be applied to latent axes or learned latent directions.

Proposition B.2 (Linear-limit reconstruction continuity)**.** *Consider a noise-aware VAE whose decoder is linear, $D_\theta(z) = P_\theta z$, and whose encoder distribution concentrates on a linear code $z = E_\phi x$. In the deterministic autoencoder limit, the data-dependent reconstruction term of the noise-aware ELBO becomes*

$$\|x - P_\theta E_\phi x\|^2_{\Sigma^{-1}}. \tag{61}$$

If the linear encoder is tied as $E_\phi = P_\theta^\dagger \Sigma^{-1}$ and $P_\theta^\dagger \Sigma^{-1} P_\theta = I_k$, this is exactly the Paper 1 Bridge Theorem objective in Eq. (6)*.*

Proof. As the encoder distribution concentrates on $E_\phi x$, the expectation of the Gaussian reconstruction term converges to the reconstruction loss evaluated at that code. Substituting the linear decoder gives $\|x - P_\theta E_\phi x\|^2_{\Sigma^{-1}}$. With the Σ^{-1}-adjoint tie, $E_\phi = P_\theta^\dagger \Sigma^{-1}$, the reconstruction map is the Σ^{-1}-orthogonal projector $P_\theta P_\theta^\dagger \Sigma^{-1}$, giving Eq. (6). The KL term remains a latent regularizer in a genuine VAE; the exact Bridge objective is recovered for the reconstruction term, or for a deterministic autoencoder limit in which that regularizer is removed, down-weighted, or held fixed. □

Status. This section is theory-ready but experimentally incomplete. The next step is to compare isotropic and noise-aware VAEs on the same controlled pulse data and check whether the metric-reversal pattern survives in nonlinear latent models. The pullback-geodesic, naturalness-energy, and factored-control diagnostics above provide the corresponding manifold-level tests, and the coverage experiment of Section 8 provides the natural stress test for whether the VAE latent geometry remains identifiable off-support. Latent coordinates themselves should be evaluated only through gauge-invariant quantities, per Section 3.4.

C Framework-Derived Extension: Noise-Aware Transformer

This appendix is a framework-derived extension, not a completed experimental claim. Read through the tuple $(\Sigma^{-1}, \mathcal{F}, P_{\text{train}} \to P_{\text{target}})$, a noise-aware Transformer is specified by: *architecture prior* — learned global pairwise attention, i.e. complete-graph message passing on prewhitened tokens (Section 3.2); *detector structure assumed* — none by default, so geometry must be *supplied* through positional/graph encodings, attention biases, or prewhitening; *coverage required* — enough data to learn the relevant pairwise structure, since the weak prior gives little off-support constraint; *failure mode* — spurious long-range fits when geometry is absent and coverage is thin; and *minimal experiment* — the prewhitening and positional-encoding ablations below, run under the coverage and architecture comparisons of Section 8. A Transformer does not automatically capture detector geometry; the rest of this appendix specifies how it can inherit the Σ^{-1} metric through prewhitening and how geometry is supplied.

C.1 Prewhitened Transformer

The safest Transformer extension is to whiten before tokenization:

$$\tilde{x} = \Sigma^{-1/2} x.$$

Once the data are whitened, Euclidean dot products in attention operate in the correct noise geometry. No modification to the attention mechanism is required for the metric itself; the noise covariance has already been absorbed into the input representation. A first ablation should therefore compare:

1. raw Transformer with MSE,
2. raw Transformer with Mahalanobis loss,
3. prewhitened Transformer with MSE in whitened space,
4. prewhitened Transformer with additional noise-aware positional encoding.

Prewhitening fixes the residual metric; it does not by itself supply locality, graph structure, or deformation stability (Section 2.2).

C.2 Time-domain, frequency-domain, and joint variants

The frequency-domain view should be read as a likelihood-coordinate choice, not as a claim that the Fourier transform separates signal from noise. For stationary noise,

$$\Sigma = F^\dagger \operatorname{diag}(S_n(f_k)) F, \qquad \Sigma^{-1} = F^\dagger \operatorname{diag}(1/S_n(f_k)) F,$$

so the FFT diagonalizes the covariance and turns the Gaussian loss into a PSD-weighted Euclidean loss over frequency bins. White noise remains present in all frequency bins; line features, band-limited noise, or colored spectra can be down-weighted or isolated only to the extent that their frequency support or PSD differs from the signal. Thus FFT/prewhitening changes the representation of the residual geometry, while the architecture still determines which temporal, channel, and artifact structure can be represented.

This distinction is useful because some physical variations become simple in frequency coordinates. A small arrival-time shift obeys

$$\hat{s}(\omega; t_0 + \tau) = e^{-i\omega\tau}\hat{s}(\omega; t_0) \approx (1 - i\omega\tau)\hat{s}(\omega; t_0),$$

so timing variation is approximately linear in the frequency representation. This explains why timing or phase variation can appear as a low-order direction in PCA, EMPCA, or a linear autoencoder. It does not imply that nonstationary artifacts, baseline excursions, channel cross-talk, or time-localized defects are well represented by frequency weighting alone.

The Transformer experiments should therefore distinguish three related but different implementations:

1. *Prewhitened or FFT-token Transformer*: tokens are formed after whitening or in a PSD-weighted frequency representation. This is the cleanest likelihood-geometry version.

2. *Time-domain Transformer with spectral late fusion*: attention operates on multi-channel time traces, while PSD, FFT, or optimal-filter summaries are injected at pooling or readout. This keeps localized temporal morphology visible while still providing spectral noise context.

3. *Dual-stream time–frequency Transformer*: one stream processes time-domain traces, another processes whitened frequency-domain features, and cross-attention or late fusion combines them.

The three versions form an ablation ladder. If the first performs best, the dominant structure is captured by likelihood-aligned spectral coordinates. If the second or third is needed, the data contain time-localized or channel-correlated structure that survives whitening and requires an architecture prior beyond frequency-domain projection.

C.3 Tokenization, embeddings, and readout

Attention acts on a set of tokens, so the first three design decisions — *what a token is*, *what each token embeds*, and *how tokens are pooled into the output* — determine which reconstruction maps the model can express. They are not interchangeable, and stating them is a precondition for any honest comparison to NFPA or a CNN.

Tokenization. For a whitened channel-time trace $\tilde{X} \in \mathbb{R}^{C \times T}$, three tokenizations are natural, and they expose different dependencies to attention:

1. *Channel tokens* (C tokens): each token embeds one channel's whitened time series. Attention then models cross-channel coupling directly, while temporal structure must be carried inside each token's embedding (e.g. by a temporal encoder).

2. *Time(-patch) tokens* ($\approx T$ tokens): each token embeds all channels at one time step or short temporal window. Attention models long-range temporal dependence; channel coupling is internal to each token.

3. *Channel-time patch tokens*: each token embeds a (channel-block $\times$ time-window) patch. Attention can reach both axes, but the token count grows and locality must be re-supplied through the patching and positional encoding.

The choice fixes the pairwise geometry attention can represent: cross-channel induction is visible to channel tokens, long-range temporal dependence to time tokens. A model cannot attend over an axis it has folded inside a token.

Embeddings. Each token typically carries additive embeddings of four kinds: (i) a *value/amplitude* embedding, a linear map of the whitened patch content; (ii) a *channel* embedding encoding sensor identity or physical coordinates (r, ϕ, z); (iii) a *time/positional* embedding encoding temporal order; and (iv) optionally a *conditioning* embedding for known context such as an energy band or noise state, which makes the model explicitly noise- or energy-conditional. The channel and positional embeddings are exactly where domain geometry enters; the ablation ladder of Section C.4 varies (ii)–(iii), including the Σ^{-1}-derived noise-spectral encoding.

Pooling and readout. The map from token representations to the task output is task-specific. Global scalar outputs (energy, event class) use a learned `CLS` token, mean pooling, or attention pooling; trace reconstruction or per-channel fields use a per-token readout. A useful detector-specific readout injects per-event optimal-filter amplitudes as local maximum-likelihood statistics into the readout (or as token features), so the network *refines* a calibrated linear estimate rather than reconstructing from scratch — the attention analogue of starting from the Paper 1 baseline.

When attention is worth its cost. Attention is justified only when the task requires nonlocal or pairwise dependencies that local or separable priors cannot express — for example cross-channel charge sharing or induction that spreads a single event across many electrodes and time samples (as in the Xenon TPC of Section 7), or explicit pairwise event geometry. If the signal is well described by a separable channel-time subspace, NFPA or a temporal CNN is a tighter hypothesis class and should be preferred; under restricted coverage the extra flexibility of attention is a liability rather than an advantage (Table 2). Attention does not "automatically solve" detector reconstruction; it pays off only when the required dependencies are genuinely nonlocal and the coverage is sufficient to learn them.

C.4 Positional-encoding ablation ladder

Because attention supplies no geometry of its own, the positional encoding is where domain geometry enters. We propose the following ablation ladder, evaluated under colored and channel-correlated noise:

1. sinusoidal PE (standard, geometry-agnostic baseline);
2. Fourier PE;
3. physical channel-position PE (sensor coordinates);
4. graph Laplacian PE from physical adjacency;
5. noise-Laplacian PE from Σ^{-1};
6. combined physical + noise PE;
7. no-PE baseline.

C.5 Noise-correlation positional encoding

The more speculative extension is to use eigenvectors of the precision operator as positional encodings. Let

$$\Sigma = V\Lambda V^\top, \qquad \Sigma^{-1} = V\Lambda^{-1}V^\top.$$

A noise-correlation positional encoding can be defined by selecting columns of V and optionally weighting them by precision eigenvalues. For stationary

noise, V is the Fourier basis and Λ^{-1} contains inverse PSD weights. Thus this proposal connects positional encoding to the same spectral signal-to-noise geometry used by the optimal filter.

Bronstein et al. (Bronstein et al., 2021, Sec. 5.4) make this connection explicit for graphs: "one may simply use the Laplacian eigenvectors of the (assumed) graph," citing Dwivedi and Bresson (2020). The noise-correlation positional encoding is an analogous precision-graph construction: after an appropriate normalization or sparsification, the noise precision Σ^{-1} can play the role of a graph operator, with the time series as the underlying graph and noise correlations as edge weights.

Proposition C.1 (Stationary-noise limit of precision-spectral PE)**.** *For stationary noise with power spectral density $S_n(f)$, the eigenvectors of the precision operator Σ^{-1} are the discrete Fourier modes, and the corresponding eigenvalues are proportional to $1/S_n(f)$. Thus the precision-spectral positional encoding reduces to Fourier positional features weighted by inverse noise PSD, the same frequency-domain weighting used by the optimal filter.*

Proof. Stationarity makes Σ circulant, up to boundary effects, so it is diagonalized by the discrete Fourier matrix F: $\Sigma = F^\dagger \mathrm{diag}(S_n(f_k))F$. Therefore $\Sigma^{-1} = F^\dagger \mathrm{diag}(1/S_n(f_k))F$. The eigenvectors are Fourier modes and the eigenvalues are the inverse PSD weights that appear in the matched-filter inner product. □

C.6 Detector-channel graph view

For multi-channel detectors, the domain Ω is often better viewed as a graph or product graph than as a single one-dimensional grid. Vertices can be time samples, readout channels, pixels, or channel-time pairs. Edges can come from physical adjacency, readout cross-talk, empirical noise correlations, or a sparsified precision matrix. One simple construction is to set edge weights $w_{ij} = |\Sigma^{-1}_{ij}|$ after thresholding weak entries, or to build a correlation graph from the normalized covariance and then use its Laplacian as an architectural prior.

This gives a detector-specific version of the geometric deep learning hierarchy. A convolutional model assumes a fixed local graph with shared filters. A Transformer uses attention, which can be interpreted as a learned data-dependent graph on the tokens. A graph neural network makes the graph explicit and performs message passing along its edges. The noise model

can enter this hierarchy in two complementary ways: prewhitening fixes the metric in token space, while the covariance or precision graph can provide an attention bias, a graph Laplacian positional encoding, or the adjacency used by a message-passing reconstruction model. This is a proposed architecture prior, not a replacement for the Σ^{-1} likelihood.

Graph-construction ablation. For the detector channel graph G, we propose comparing five constructions:

1. physical adjacency graph;
2. covariance graph (normalized empirical covariance);
3. precision graph with weights $|\Sigma^{-1}_{ij}|$ after thresholding;
4. learned attention graph;
5. complete graph (no structural prior).

Evaluation metrics: weighted reconstruction residual, energy resolution, position error, equivariance error under graph relabeling (Eq. (50)), and robustness under channel dropout and noise perturbation. Per Section 3.2, relabeling equivariance requires that adjacency, coordinates, and covariance be relabeled consistently; it does not assume physical exchangeability of channels.

C.7 Local-linear diagnostics for detector Transformers

Recent work on local linearity in Transformer language models (Skifstad et al., 2026) suggests a useful diagnostic program, by analogy only. We do not import the LQR controller or claim that detector Transformers have been validated as locally linear. The transferable idea is narrower: a trained Transformer block can be probed through local Jacobians, and the stability of the dominant Jacobian subspaces can be tested across physically meaningful regions of the detector manifold.

Write the hidden representation after layer ℓ as z_ℓ and the block map as

$$z_{\ell+1} = \phi_\ell(z_\ell).$$

Around a reachable detector state $z^\star_\ell$, the first-order model is

$$z_{\ell+1} \approx \phi_\ell(z^\star_\ell) + A^\star_\ell(z_\ell - z^\star_\ell), \qquad A^\star_\ell = \left.\frac{\partial \phi_\ell}{\partial z_\ell}\right|_{z^\star_\ell}.$$

The detector question is not whether this linearization is a global model, but whether its dominant modes are stable across strata in energy, arrival time, position, channel-sharing, pulse shape, artifact class, and noise state. Let $Q_{\ell,k}(z)$ be an orthonormal basis for the top k right singular vectors of $A_\ell(z)$. A gauge-safe alignment statistic is

$$a_{\ell,k}(z, z') = \frac{1}{k} \left\| Q_{\ell,k}(z)^\top Q_{\ell,k}(z') \right\|_F^2 ,$$

which equals one for identical k-dimensional subspaces and decreases as the dominant local modes rotate. Reporting this statistic within and across detector strata tests whether the Transformer uses stable physical modes or sample-specific nuisance directions.

Table 10: Local-linear diagnostics for a trained detector Transformer. These are interpretability and validation tests, not evidence that attention should be preferred before the ablations are run.

Diagnostic	Detector interpretation
Singular spectrum of A_ℓ	Hidden directions that dominate layer-to-layer propagation
Top-subspace alignment across event strata	Stable physical modes versus sample-specific artifacts
Alignment with OF / EMPCA / NFPA directions	Whether the nonlinear model reuses known likelihood-aligned modes
Alignment across PSD or noise states	Robustness versus noise-state dependence
Sensitivity to PSD, channel, or dropout perturbations	Which detector design and readout features control the reconstruction
Activation interventions along candidate directions	Whether an interpretable hidden direction causally moves the intended output

The same framework gives causal intervention tests. For a detector factor $h \in \{E, t_0, r, \phi, \text{baseline}, \text{artifact}, \text{noise}\}$, define a contrastive hidden direction, for example,

$$v_{\ell,h} \propto \mathbb{E}[z_\ell \mid h + \Delta h] - \mathbb{E}[z_\ell \mid h],$$

with the conditioning implemented by matched simulations, calibration samples, or controlled injections. Perturbing the hidden state as $z_\ell \mapsto z_\ell + \alpha v_{\ell,h}$ and running the remaining layers tests whether the output changes in the

expected physical coordinate while unrelated coordinates remain stable. A timing direction should move t_0 more than energy; an artifact direction should affect residual or veto behavior without creating a spurious physical signal. These tests are gauge-safe because they evaluate input-output sensitivity and subspaces, not the raw coordinate system of the hidden representation.

Status. This section is a proposal. It is motivated by the geometric-deep-learning view that positional encodings should reflect the domain geometry, but it should not be claimed to outperform standard encodings before the ablation is run. The appropriate first tests are the metric ablation, the coverage experiment, and the architecture-ladder comparison of Section 8, followed by the time–frequency, graph-construction, positional-encoding, and local-linear diagnostics specified above.

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